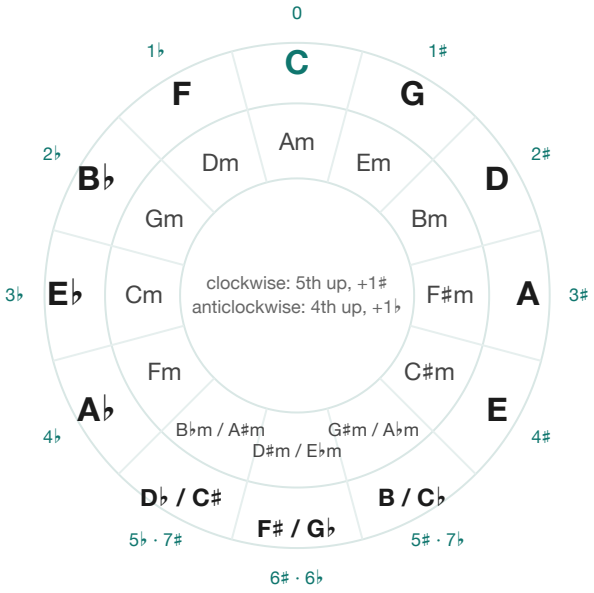
GUITAR CARDS · MEMORY PACK

Memory Essentials

The facts to recall instantly, with no instrument: every key, every mode, functional harmony, guide tones and superimposition — in all 12 keys.



A Keys
Intervals, circle of fifths, every note in every key, the degree triangle

B Scales and modes
Major, harmonic minor and melodic minor modes; diminished and whole tone

C Functional harmony
Secondary dominants, tritone subs, modal interchange

D Guide tones
3rds and 7ths, the ii–V–I rule, canonical lines

E Superimposition
Which triad or pentatonic to play over each chord, and the tensions it adds

The printed companion to the Memory pack. Every note name in these tables is generated by the same spelling engine that writes the cards, so each letter appears once per scale and chord (E♯ in F♯ major, C♭ in D♭7, B𝄫 in C°7). Where a root needs a double accidental, the plain enharmonic is shown beside it in grey. Fretboard shapes live in the Guitar Mastery guide; this booklet is facts and tables only.

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- **Start with the rules page.** Every table is one rule applied to every key; recite the rule, then check yourself against the table.
- **Accent colour** marks the note that matters: the newest accidental, the raised note, the characteristic degree of a mode, the out-of-key note of a secondary dominant.
- **Keys run in circle order** (C G D A E B F♯ D♭ A♭ E♭ B♭ F) so neighbouring rows differ by one note. Cover a row, say it, uncover.
- Roman numerals are relative to the **major** scale of the tonic: a minor key reads i iiø7 ♭III iv v ♭VI ♭VII.

FRONT

Rules that generate everything

Learn these lines first. Each one regenerates a whole table in the pages that follow.

A Keys
Intervals, circle of fifths, every note in every key, the degree triangle

A1 **Intervals (reference).** Spell the root's major scale; major and perfect intervals are its notes. One semitone smaller than major is minor; smaller than perfect is diminished; larger is augmented.

A2 **Circle of fifths & key signatures.** Clockwise: up a 5th, add a sharp (the new key's 7th). Anticlockwise: up a 4th, add a flat (the new key's 4th). Sharp keys: up a semitone from the last sharp. Flat keys: the second-to-last flat.

A3 **Relative and parallel.** Relative minor: same signature, starts on the 6th (a minor 3rd below the major tonic). Parallel minor: same tonic.

A4 **Every note in every key.** Major keys in circle order, each one note different from the last; each letter once. Natural minor = relative major from its 6th. Harmonic minor = natural minor with the 7th raised. Melodic minor = major with a ♭3.

A5 **The degree triangle.** Major: maj7, m7, m7, maj7, 7, m7, m7♭5 — Ionian, Dorian, Phrygian, Lydian, Mixolydian, Aeolian, Locrian. Natural minor is the same cycle from the 6th: i m7, iiø7, ♭IIImaj7, iv m7, v m7, ♭VImaj7, ♭VII7.

B Scales and modes
Major, harmonic minor and melodic minor modes; diminished and whole tone

B1 **Major modes: the brightness ladder.** Lydian ♯4 · Ionian · Mixolydian ♭7 · Dorian ♭3 · Aeolian ♭6 · Phrygian ♭2 · Locrian ♭5 — each darkens one more note. Parents: Dorian a whole step down, Phrygian a M3 down, Lydian a 4th down, Mixolydian a 5th down, Aeolian a m3 up, Locrian a semitone up.

B2 **Harmonic minor: one raised note.** Each harmonic-minor mode is a major mode on the same root with one note raised (Ultralocrian: Locrian with ♭4 and 𝄫7). Phrygian dominant = harmonic minor a 4th above.

B3 **Melodic minor: one raised note.** Each melodic-minor mode is a major mode with one note changed. X7alt = melodic minor a semitone above X; X7♯11 = a 5th above; Xm7♭5 = a minor 3rd above.

B4 **Symmetric scales.** Half-whole: H W repeating, over 7♭9 (repeats every minor 3rd). Whole-half: W H, over dim7. Whole tone: all whole steps, over 7♯5 (no perfect 5th).

C Functional harmony
Secondary dominants, tritone subs, modal interchange

C1 **Secondary dominants.** Go up a 5th from the target and make it a dominant 7th. Its 3rd is the target's leading tone (V7/IV adds the ♭7 instead).

C2 **Tritone substitution.** The dominant a tritone away has the same 3rd and 7th, swapped, and resolves down a semitone.

C3 **Modal interchange.** Borrow from the PARALLEL modes (same tonic). Flat-side chords come from darker modes: ♭VI needs a ♭6, so Aeolian or darker.

C4 **Modal interchange in context.** Each borrowed chord is a move with a job: name its numerals, its source mode and the one voice that moves (usually ♭6 → 5 or ♭7 → 1).

D Guide tones
3rds and 7ths, the ii–V–I rule, canonical lines

D1 **Guide tones of any chord.** The 3rd says major or minor; the 7th says major 7th, dominant or diminished. Only the dominant has a tritone between them.

D2 **The ii–V–I rule.** Roots down a 5th: the 7th falls a step to the next 3rd; the 3rd holds to become the next 7th.

D3 **The canonical progressions.** 7th falls, 3rd holds (roots down a 5th). Tritone sub: same lines, only the root moves. Cycle of dominants: two chromatic lines falling. Blues I7–IV7: both fall a semitone.

E Superimposition
Which triad or pentatonic to play over each chord, and the tensions it adds

E1 **Triads over chords.** Name the triad by the chord degree it sits on, then read each note as a tension over the chord root. Over a dominant: II = 9 ♯11 13, VI = 13 ♭9 3, ♭VI = ♭13 1 ♯9, ♭V = ♯11 ♭7 ♭9.

E2 **Pentatonics over chords.** Pentatonics a step or a 5th up stay inside (Dorian, Lydian, Mixolydian colours); the minor pentatonic a minor 3rd above a dominant is the altered sound.

Seventeen rules, one per unit of the Memory pack. Everything else in this booklet is one of these rules applied to all 12 keys: if a table cell is ever in doubt, rebuild it from the rule.

A1 Intervals

A1 RULE Spell the root's major scale; major and perfect intervals are its notes. One semitone smaller than major is minor; smaller than perfect is diminished; larger is augmented.

The major scale is the measuring stick. Measured up from the root, every note of a major scale is a **major or perfect** interval: M2, M3, P4, P5, M6, M7. One semitone smaller than major is **minor**; one smaller than perfect is **diminished**; one larger than either is **augmented**.

So to find any interval above a note, spell that note's major scale and adjust. A minor 6th above E♭ is the 6th of E♭ major (C) lowered a semitone: **C♭**. Not B, because a 6th must land on the sixth letter.

Hold these: inversions add to 9 (M3 inverts to m6, P4 to P5), and major inverts to minor. Extensions are the 2nd, 4th and 6th plus an octave: 9 = 2, 11 = 4, 13 = 6.

The eleven intervals

Interval	Name	Semitones	Degree	Inverts to	Inverted semitones
m2	minor 2nd	1	♭2	M7	11
M2	major 2nd	2	2	m7	10
m3	minor 3rd	3	♭3 / ♯2	M6	9
M3	major 3rd	4	3	m6	8
P4	perfect 4th	5	4	P5	7
tritone	tritone	6	♯4 / ♭5	tritone	6
P5	perfect 5th	7	5	P4	5
m6	minor 6th	8	♭6 / ♯5	M3	4
M6	major 6th	9	6	m3	3
m7	minor 7th	10	♭7	M2	2
M7	major 7th	11	7	m2	1
P8	octave	12	8	P1	0

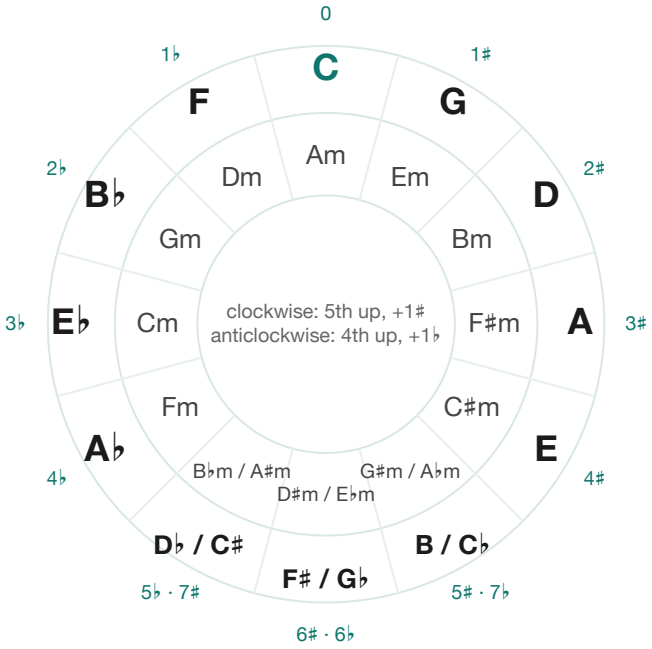
Extensions = 2nd, 4th, 6th an octave up

Extension	Same note as	Semitones above root (mod 12)
♭9	♭2	1
9	2	2
♯9	♯2 (sounds as ♭3)	3
11	4	5
♯11	♯4 (sounds as ♭5)	6
♭13	♭6 (sounds as ♯5)	8
13	6	9

> **Hold these:** inversions add to **9** (M3 ↔ m6, P4 ↔ P5) and their semitones add to **12**; major ↔ minor, perfect stays perfect, the tritone stays a tritone. Extensions: **9 = 2, 11 = 4, 13 = 6**.

A2 Circle of fifths and key signatures

A2 RULE Clockwise: up a 5th, add a sharp (the new key's 7th). Anticlockwise: up a 4th, add a flat (the new key's 4th). Sharp keys: up a semitone from the last sharp. Flat keys: the second-to-last flat.



Going **clockwise** moves up a 5th and adds one sharp. The new sharp is always the new key's **7th degree**, raised so it leads to the tonic. Going **anticlockwise** moves up a 4th and adds one flat, which is always the new key's **4th degree**.

Order of sharps: **F C G D A E B** ("Father Charles Goes Down And Ends Battle"). Order of flats: the same backwards.

Two shortcuts name a key from its signature:

- **Sharp keys:** go up a semitone from the last sharp. Four sharps end on D♯, so the key is E.
- **Flat keys:** the second-to-last flat names the key. Four flats are B♭ E♭ A♭ D♭, so the key is A♭. (F, with one flat, is the one to memorise.)

Key signatures: all 15 spellings

Major key	Sig.	Accidentals, in order	Newest accidental	Rel. minor	Naming shortcut
C	—	—	—	Am	—
G	1♯	F♯	F♯ = 7th of G	Em	last sharp F♯ + semitone = G
D	2♯	F♯ C♯	C♯ = 7th of D	Bm	last sharp C♯ + semitone = D
A	3♯	F♯ C♯ G♯	G♯ = 7th of A	F♯m	last sharp G♯ + semitone = A
E	4♯	F♯ C♯ G♯ D♯	D♯ = 7th of E	C♯m	last sharp D♯ + semitone = E
B	5♯	F♯ C♯ G♯ D♯ A♯	A♯ = 7th of B	G♯m	last sharp A♯ + semitone = B
F♯	6♯	F♯ C♯ G♯ D♯ A♯ E♯	E♯ = 7th of F♯	D♯m	last sharp E♯ + semitone = F♯
C♯	7♯	F♯ C♯ G♯ D♯ A♯ E♯ B♯	B♯ = 7th of C♯	A♯m	last sharp B♯ + semitone = C♯
F	1♭	B♭	B♭ = 4th of F	Dm	one flat: memorise F
B♭	2♭	B♭ E♭	E♭ = 4th of B♭	Gm	second-to-last flat: B♭
E♭	3♭	B♭ E♭ A♭	A♭ = 4th of E♭	Cm	second-to-last flat: E♭
A♭	4♭	B♭ E♭ A♭ D♭	D♭ = 4th of A♭	Fm	second-to-last flat: A♭
D♭	5♭	B♭ E♭ A♭ D♭ G♭	G♭ = 4th of D♭	B♭m	second-to-last flat: D♭
G♭	6♭	B♭ E♭ A♭ D♭ G♭ C♭	C♭ = 4th of G♭	E♭m	second-to-last flat: G♭
C♭	7♭	B♭ E♭ A♭ D♭ G♭ C♭ F♭	F♭ = 4th of C♭	A♭m	second-to-last flat: C♭

Order of sharps: F C G D A E B · **Order of flats:** B E A D G C F (the same backwards).

Enharmonic pairs on the circle: B = C♭ (5♯ / 7♭), F♯ = G♭ (6♯ / 6♭), C♯ = D♭ (7♯ / 5♭). Sharps + flats of the pair always add to 12.

A3 Relative and parallel keys

A3 RULE Relative minor: same signature, starts on the 6th (a minor 3rd below the major tonic). Parallel minor: same tonic.

The **relative** minor shares the key signature and starts on the 6th degree, a minor 3rd below the major tonic (C major and A minor). The **parallel** minor shares the tonic (C major and C minor).

Modal interchange (Module C) borrows from **parallel** keys, so the distinction matters.

Major key	Sig.	Relative minor (6th)	Parallel minor (same tonic)	Parallel minor sig.	Its relative major (♭III)	Clockwise neighbour (5th up)	Anticlockwise neighbour (4th up)
C	—	Am	Cm	3♭	E♭	G	F
G	1♯	Em	Gm	2♭	B♭	D	C
D	2♯	Bm	Dm	1♭	F	A	G
A	3♯	F♯m	Am	—	C	E	D
E	4♯	C♯m	Em	1♯	G	B	A
B	5♯	G♯m	Bm	2♯	D	F♯	E
F♯	6♯	D♯m	F♯m	3♯	A	C♯	B
D♭	5♭	B♭m	C♯m (enh.)	4♯	E	A♭	G♭
A♭	4♭	Fm	A♭m	7♭	C♭	E♭	D♭
E♭	3♭	Cm	E♭m	6♭	G♭	B♭	A♭
B♭	2♭	Gm	B♭m	5♭	D♭	F	E♭
F	1♭	Dm	Fm	4♭	A♭	C	B♭

Parallel minor = three more flats (or three fewer sharps) than the major on the same tonic. Relative minor = down a minor 3rd; relative major = up a minor 3rd.

From the minor side

Minor key	Sig.	Relative major (♭3)	Parallel major (same tonic)	Parallel major sig.	Dominant (V7, harmonic minor)	Leading tone (raised 7th)
Am	—	C	A	3♯	E7	G♯
Em	1♯	G	E	4♯	B7	D♯
Bm	2♯	D	B	5♯	F♯7	A♯
F♯m	3♯	A	F♯	6♯	C♯7	E♯
C♯m	4♯	E	C♯	7♯	G♯7	B♯
G♯m	5♯	B	A♭ (enh.)	4♭	D♯7	F𝄪
D♯m	6♯	F♯	E♭ (enh.)	3♭	A♯7	C𝄪
B♭m	5♭	D♭	B♭	2♭	F7	A
Fm	4♭	A♭	F	1♭	C7	E
Cm	3♭	E♭	C	—	G7	B
Gm	2♭	B♭	G	1♯	D7	F♯
Dm	1♭	F	D	2♯	A7	C♯

A4 Every note in every key

A4 RULE Major keys in circle order, each one note different from the last; each letter once. Natural minor = relative major from its 6th. Harmonic minor = natural minor with the 7th raised. Melodic minor = major with a ♭3.

Learn major keys in circle order (C, G, D, A, E, B, F♯, then F, B♭, E♭, A♭, D♭, G♭), so each key adds one note change to the last. Each letter appears once, which is why F♯ major has **E♯**, not F.

The three minors are built from things you know:

- **Natural minor:** the relative major started from its 6th. E minor is G major from E.
- **Harmonic minor:** natural minor with the 7th raised. E F♯ G A B C **D♯**.
- **Melodic minor (jazz form):** major with a ♭3. E F♯ **G** A B C♯ D♯.

Major keys (circle order)

Key	Sig.	1	2	3	4	5	6	7
C	—	C	D	E	F	G	A	B
G	1♯	G	A	B	C	D	E	**F♯**
D	2♯	D	E	F♯	G	A	B	**C♯**
A	3♯	A	B	C♯	D	E	F♯	**G♯**
E	4♯	E	F♯	G♯	A	B	C♯	**D♯**
B	5♯	B	C♯	D♯	E	F♯	G♯	**A♯**
F♯	6♯	F♯	G♯	A♯	B	C♯	D♯	**E♯**
D♭	5♭	D♭	E♭	F	**G♭**	A♭	B♭	C
A♭	4♭	A♭	B♭	C	**D♭**	E♭	F	G
E♭	3♭	E♭	F	G	**A♭**	B♭	C	D
B♭	2♭	B♭	C	D	**E♭**	F	G	A
F	1♭	F	G	A	**B♭**	C	D	E

Highlighted: the accidental this key adds to its neighbour (a new sharp is the 7th; a new flat is the 4th).

Natural minor (relative major from its 6th)

Tonic	Rel. major	1	2	♭3	4	5	♭6	♭7
A	C	A	B	C	D	E	F	G
E	G	E	F♯	G	A	B	C	D
B	D	B	C♯	D	E	F♯	G	A
F♯	A	F♯	G♯	A	B	C♯	D	E
C♯	E	C♯	D♯	E	F♯	G♯	A	B
G♯	B	G♯	A♯	B	C♯	D♯	E	F♯
D♯	F♯	D♯	E♯	F♯	G♯	A♯	B	C♯
B♭	D♭	B♭	C	D♭	E♭	F	G♭	A♭
F	A♭	F	G	A♭	B♭	C	D♭	E♭
C	E♭	C	D	E♭	F	G	A♭	B♭
G	B♭	G	A	B♭	C	D	E♭	F
D	F	D	E	F	G	A	B♭	C

Harmonic minor (natural minor, 7th raised)

Tonic	Rel. major	1	2	♭3	4	5	♭6	7
A	C	A	B	C	D	E	F	**G♯**
E	G	E	F♯	G	A	B	C	**D♯**
B	D	B	C♯	D	E	F♯	G	**A♯**
F♯	A	F♯	G♯	A	B	C♯	D	**E♯**
C♯	E	C♯	D♯	E	F♯	G♯	A	**B♯**
G♯	B	G♯	A♯	B	C♯	D♯	E	**F𝄪**
D♯	F♯	D♯	E♯	F♯	G♯	A♯	B	**C𝄪**
B♭	D♭	B♭	C	D♭	E♭	F	G♭	**A**
F	A♭	F	G	A♭	B♭	C	D♭	**E**
C	E♭	C	D	E♭	F	G	A♭	**B**
G	B♭	G	A	B♭	C	D	E♭	**F♯**
D	F	D	E	F	G	A	B♭	**C♯**

Highlighted: the raised 7th, the leading tone that gives minor its V7.

Melodic minor, jazz form (major with ♭3)

Tonic	Rel. major	1	2	♭3	4	5	6	7
A	C	A	B	C	D	E	**F♯**	**G♯**
E	G	E	F♯	G	A	B	**C♯**	**D♯**
B	D	B	C♯	D	E	F♯	**G♯**	**A♯**
F♯	A	F♯	G♯	A	B	C♯	**D♯**	**E♯**
C♯	E	C♯	D♯	E	F♯	G♯	**A♯**	**B♯**
G♯	B	G♯	A♯	B	C♯	D♯	**E♯**	**F𝄪**
D♯	F♯	D♯	E♯	F♯	G♯	A♯	**B♯**	**C𝄪**
B♭	D♭	B♭	C	D♭	E♭	F	**G**	**A**
F	A♭	F	G	A♭	B♭	C	**D**	**E**
C	E♭	C	D	E♭	F	G	**A**	**B**
G	B♭	G	A	B♭	C	D	**E**	**F♯**
D	F	D	E	F	G	A	**B**	**C♯**

Highlighted: the raised 6th and 7th compared with natural minor.

A5 The degree triangle in all 24 keys

A5 RULE Major: maj7, m7, m7, maj7, 7, m7, m7♭5 — Ionian, Dorian, Phrygian, Lydian, Mixolydian, Aeolian, Locrian. Natural minor is the same cycle from the 6th: i m7, iiø7, ♭IIImaj7, iv m7, v m7, ♭VImaj7, ♭VII7.

Stack 7th chords on each note of a major scale and you always get the same pattern: **maj7, m7, m7, maj7, 7, m7, m7♭5**. Memorise it as a rhythm ("major, minor, minor, major, dominant, minor, half-diminished"). The mode on each degree is in the same order: **Ionian, Dorian, Phrygian, Lydian, Mixolydian, Aeolian, Locrian**.

Natural minor is the same cycle started from the 6th, so it reads m7, m7♭5, maj7, m7, m7, maj7, 7 on **i, ii, ♭III, iv, v, ♭VI, ♭VII**.

Each degree gives three answers at once: the note, the chord quality and the mode. E minor, ♭VII is **D, D7, Mixolydian**.

Major keys: the seven 7th chords and their modes

Key	Imaj7 Ionian	ii7 Dorian	iii7 Phrygian	IVmaj7 Lydian	V7 Mixolydian	vi7 Aeolian	viiø7 Locrian
C	Cmaj7	Dm7	Em7	Fmaj7	G7	Am7	Bm7♭5
G	Gmaj7	Am7	Bm7	Cmaj7	D7	Em7	F♯m7♭5
D	Dmaj7	Em7	F♯m7	Gmaj7	A7	Bm7	C♯m7♭5
A	Amaj7	Bm7	C♯m7	Dmaj7	E7	F♯m7	G♯m7♭5
E	Emaj7	F♯m7	G♯m7	Amaj7	B7	C♯m7	D♯m7♭5
B	Bmaj7	C♯m7	D♯m7	Emaj7	F♯7	G♯m7	A♯m7♭5
F♯	F♯maj7	G♯m7	A♯m7	Bmaj7	C♯7	D♯m7	E♯m7♭5
D♭	D♭maj7	E♭m7	Fm7	G♭maj7	A♭7	B♭m7	Cm7♭5
A♭	A♭maj7	B♭m7	Cm7	D♭maj7	E♭7	Fm7	Gm7♭5
E♭	E♭maj7	Fm7	Gm7	A♭maj7	B♭7	Cm7	Dm7♭5
B♭	B♭maj7	Cm7	Dm7	E♭maj7	F7	Gm7	Am7♭5
F	Fmaj7	Gm7	Am7	B♭maj7	C7	Dm7	Em7♭5

Natural minor keys: i · iiø7 · ♭III · iv · v · ♭VI · ♭VII7

Key	i7 Aeolian	iiø7 Locrian	♭IIImaj7 Ionian	iv7 Dorian	v7 Phrygian	♭VImaj7 Lydian	♭VII7 Mixolydian
Am	Am7	Bm7♭5	Cmaj7	Dm7	Em7	Fmaj7	G7
Em	Em7	F♯m7♭5	Gmaj7	Am7	Bm7	Cmaj7	D7
Bm	Bm7	C♯m7♭5	Dmaj7	Em7	F♯m7	Gmaj7	A7
F♯m	F♯m7	G♯m7♭5	Amaj7	Bm7	C♯m7	Dmaj7	E7
C♯m	C♯m7	D♯m7♭5	Emaj7	F♯m7	G♯m7	Amaj7	B7
G♯m	G♯m7	A♯m7♭5	Bmaj7	C♯m7	D♯m7	Emaj7	F♯7
D♯m	D♯m7	E♯m7♭5	F♯maj7	G♯m7	A♯m7	Bmaj7	C♯7
B♭m	B♭m7	Cm7♭5	D♭maj7	E♭m7	Fm7	G♭maj7	A♭7
Fm	Fm7	Gm7♭5	A♭maj7	B♭m7	Cm7	D♭maj7	E♭7
Cm	Cm7	Dm7♭5	E♭maj7	Fm7	Gm7	A♭maj7	B♭7
Gm	Gm7	Am7♭5	B♭maj7	Cm7	Dm7	E♭maj7	F7
Dm	Dm7	Em7♭5	Fmaj7	Gm7	Am7	B♭maj7	C7

Harmonic minor keys: where the V7 and vii°7 come from

Key	i(maj7) Harmonic minor	iiø7 Locrian ♮6	♭IIImaj7♯5 Ionian ♯5	iv7 Dorian ♯4	V7 Phrygian dominant	♭VImaj7 Lydian ♯2	vii°7 Ultralocrian
Am	AmMaj7	Bm7♭5	Cmaj7♯5	Dm7	E7	Fmaj7	G♯°7
Em	EmMaj7	F♯m7♭5	Gmaj7♯5	Am7	B7	Cmaj7	D♯°7
Bm	BmMaj7	C♯m7♭5	Dmaj7♯5	Em7	F♯7	Gmaj7	A♯°7
F♯m	F♯mMaj7	G♯m7♭5	Amaj7♯5	Bm7	C♯7	Dmaj7	E♯°7
C♯m	C♯mMaj7	D♯m7♭5	Emaj7♯5	F♯m7	G♯7	Amaj7	B♯°7
G♯m	G♯mMaj7	A♯m7♭5	Bmaj7♯5	C♯m7	D♯7	Emaj7	F𝄪°7 = G°7
D♯m	D♯mMaj7	E♯m7♭5	F♯maj7♯5	G♯m7	A♯7	Bmaj7	C𝄪°7 = D°7
B♭m	B♭mMaj7	Cm7♭5	D♭maj7♯5	E♭m7	F7	G♭maj7	A°7
Fm	FmMaj7	Gm7♭5	A♭maj7♯5	B♭m7	C7	D♭maj7	E°7
Cm	CmMaj7	Dm7♭5	E♭maj7♯5	Fm7	G7	A♭maj7	B°7
Gm	GmMaj7	Am7♭5	B♭maj7♯5	Cm7	D7	E♭maj7	F♯°7
Dm	DmMaj7	Em7♭5	Fmaj7♯5	Gm7	A7	B♭maj7	C♯°7

Melodic minor keys: the jazz-minor chords

Key	i(maj7) Melodic minor	ii7 Dorian ♭2	♭IIImaj7♯5 Lydian augmented	IV7 Lydian dominant	V7 Mixolydian ♭6	viø7 Locrian ♮2	VII7alt Altered
Am	AmMaj7	Bm7	Cmaj7♯5	D7	E7	F♯m7♭5	G♯7alt
Em	EmMaj7	F♯m7	Gmaj7♯5	A7	B7	C♯m7♭5	D♯7alt
Bm	BmMaj7	C♯m7	Dmaj7♯5	E7	F♯7	G♯m7♭5	A♯7alt
F♯m	F♯mMaj7	G♯m7	Amaj7♯5	B7	C♯7	D♯m7♭5	E♯7alt
C♯m	C♯mMaj7	D♯m7	Emaj7♯5	F♯7	G♯7	A♯m7♭5	B♯7alt
G♯m	G♯mMaj7	A♯m7	Bmaj7♯5	C♯7	D♯7	E♯m7♭5	F𝄪7alt = G7alt
D♯m	D♯mMaj7	E♯m7	F♯maj7♯5	G♯7	A♯7	B♯m7♭5	C𝄪7alt = D7alt
B♭m	B♭mMaj7	Cm7	D♭maj7♯5	E♭7	F7	Gm7♭5	A7alt
Fm	FmMaj7	Gm7	A♭maj7♯5	B♭7	C7	Dm7♭5	E7alt
Cm	CmMaj7	Dm7	E♭maj7♯5	F7	G7	Am7♭5	B7alt
Gm	GmMaj7	Am7	B♭maj7♯5	C7	D7	Em7♭5	F♯7alt
Dm	DmMaj7	Em7	Fmaj7♯5	G7	A7	Bm7♭5	C♯7alt

Where a chord lives: the major keys that contain it

Go down from the chord root by its degree: ii → a step down, iii → a major 3rd down, IV → a 4th down, V → a 5th down, vi → a major 6th down, vii → a semitone up. A minor 7 lives in three keys, a maj7 in two, a dominant 7 and a m7♭5 in one.

m7 is ii · iii · vi of…	maj7 is I · IV of…	7 is V of…	m7♭5 is vii of…
Cm7: B♭ · A♭ · E♭	**Cmaj7**: C · G	**C7**: F	**Cm7♭5**: D♭
Gm7: F · E♭ · B♭	**Gmaj7**: G · D	**G7**: C	**Gm7♭5**: A♭
Dm7: C · B♭ · F	**Dmaj7**: D · A	**D7**: G	**Dm7♭5**: E♭
Am7: G · F · C	**Amaj7**: A · E	**A7**: D	**Am7♭5**: B♭
Em7: D · C · G	**Emaj7**: E · B	**E7**: A	**Em7♭5**: F
Bm7: A · G · D	**Bmaj7**: B · F♯	**B7**: E	**Bm7♭5**: C
F♯m7: E · D · A	**F♯maj7**: F♯ · C♯	**F♯7**: B	**F♯m7♭5**: G
C♯m7: B · A · E	**C♯maj7**: C♯ · G♯	**C♯7**: F♯	**C♯m7♭5**: D
G♯m7: F♯ · E · B	**A♭maj7**: A♭ · E♭	**A♭7**: D♭	**G♯m7♭5**: A
D♯m7: C♯ · B · F♯	**E♭maj7**: E♭ · B♭	**E♭7**: A♭	**D♯m7♭5**: E
B♭m7: A♭ · G♭ · D♭	**B♭maj7**: B♭ · F	**B♭7**: E♭	**A♯m7♭5**: B
Fm7: E♭ · D♭ · A♭	**Fmaj7**: F · C	**F7**: B♭	**Fm7♭5**: G♭

B1 Major modes: the brightness ladder

B1 RULE Lydian ♯4 · Ionian · Mixolydian ♭7 · Dorian ♭3 · Aeolian ♭6 · Phrygian ♭2 · Locrian ♭5 — each darkens one more note. Parents: Dorian a whole step down, Phrygian a M3 down, Lydian a 4th down, Mixolydian a 5th down, Aeolian a m3 up, Locrian a semitone up.

Every mode can be written as the major scale with some notes flattened (or, for Lydian, one note sharpened). Ordered from brightest to darkest, **each mode darkens exactly one more note than the last**. Learn the ladder and you can spell any mode on any root: spell the major scale, then apply the alterations.

The second way in is the **parent scale**. Each mode is a major scale started on a different degree: Dorian's parent is a whole step down, Phrygian's a major 3rd down, Lydian's a 4th down, Mixolydian's a 5th down, Aeolian's a minor 3rd up and Locrian's a semitone up. E Phrygian is C major from E. Use the ladder for sound and the parent for spelling speed.

Worked example. Spell A Mixolydian. Ladder: A major is A B C♯ D E F♯ G♯; flatten the 7th: A B C♯ D E F♯ G. Parent check: a 5th below A is D, and D major from A gives the same notes.

The ladder, brightest to darkest

#	Mode	Formula	Changed note	7th chord	Colour	Degree of major	Parent major
1	**Lydian**	1 2 3 ♯4 5 6 7	♯4	maj7	♯11	IV	a 4th down
2	**Ionian**	1 2 3 4 5 6 7	—	maj7	9	I	same
3	**Mixolydian**	1 2 3 4 5 6 ♭7	♭7	7	9, 13	V	a 5th down
4	**Dorian**	1 2 ♭3 4 5 6 ♭7	♭3	m7	♮13	ii	a whole step down
5	**Aeolian**	1 2 ♭3 4 5 ♭6 ♭7	♭6	m7	♭13	vi	a minor 3rd up
6	**Phrygian**	1 ♭2 ♭3 4 5 ♭6 ♭7	♭2	m7	♭9	iii	a major 3rd down
7	**Locrian**	1 ♭2 ♭3 4 ♭5 ♭6 ♭7	♭5	m7♭5	♭9	vii	a semitone up

All seven major modes on all 12 roots

Rows in circle order; each root spelled with the fewest accidentals. Highlighted column: the note that mode changes on the ladder. Last column: the major scale with the same notes.

Lydian major with ♯4 · parent a 4th down

Root	1	2	3	♯4	5	6	7	7th chord	Parent major
C	C	D	E	**F♯**	G	A	B	Cmaj7	G
G	G	A	B	**C♯**	D	E	F♯	Gmaj7	D
D	D	E	F♯	**G♯**	A	B	C♯	Dmaj7	A
A	A	B	C♯	**D♯**	E	F♯	G♯	Amaj7	E
E	E	F♯	G♯	**A♯**	B	C♯	D♯	Emaj7	B
B	B	C♯	D♯	**E♯**	F♯	G♯	A♯	Bmaj7	F♯
G♭	G♭	A♭	B♭	**C**	D♭	E♭	F	G♭maj7	D♭
D♭	D♭	E♭	F	**G**	A♭	B♭	C	D♭maj7	A♭
A♭	A♭	B♭	C	**D**	E♭	F	G	A♭maj7	E♭
E♭	E♭	F	G	**A**	B♭	C	D	E♭maj7	B♭
B♭	B♭	C	D	**E**	F	G	A	B♭maj7	F
F	F	G	A	**B**	C	D	E	Fmaj7	C

Ionian the major scale · parent same

Root	1	2	3	4	5	6	7	7th chord	Parent major
C	C	D	E	F	G	A	B	Cmaj7	C
G	G	A	B	C	D	E	F♯	Gmaj7	G
D	D	E	F♯	G	A	B	C♯	Dmaj7	D
A	A	B	C♯	D	E	F♯	G♯	Amaj7	A
E	E	F♯	G♯	A	B	C♯	D♯	Emaj7	E
B	B	C♯	D♯	E	F♯	G♯	A♯	Bmaj7	B
F♯	F♯	G♯	A♯	B	C♯	D♯	E♯	F♯maj7	F♯
D♭	D♭	E♭	F	G♭	A♭	B♭	C	D♭maj7	D♭
A♭	A♭	B♭	C	D♭	E♭	F	G	A♭maj7	A♭
E♭	E♭	F	G	A♭	B♭	C	D	E♭maj7	E♭
B♭	B♭	C	D	E♭	F	G	A	B♭maj7	B♭
F	F	G	A	B♭	C	D	E	Fmaj7	F

Mixolydian major with ♭7 · parent a 5th down

Root	1	2	3	4	5	6	♭7	7th chord	Parent major
C	C	D	E	F	G	A	**B♭**	C7	F
G	G	A	B	C	D	E	**F**	G7	C
D	D	E	F♯	G	A	B	**C**	D7	G
A	A	B	C♯	D	E	F♯	**G**	A7	D
E	E	F♯	G♯	A	B	C♯	**D**	E7	A
B	B	C♯	D♯	E	F♯	G♯	**A**	B7	E
F♯	F♯	G♯	A♯	B	C♯	D♯	**E**	F♯7	B
C♯	C♯	D♯	E♯	F♯	G♯	A♯	**B**	C♯7	F♯
A♭	A♭	B♭	C	D♭	E♭	F	**G♭**	A♭7	D♭
E♭	E♭	F	G	A♭	B♭	C	**D♭**	E♭7	A♭
B♭	B♭	C	D	E♭	F	G	**A♭**	B♭7	E♭
F	F	G	A	B♭	C	D	**E♭**	F7	B♭

Dorian major with ♭3 · parent a whole step down

Root	1	2	♭3	4	5	6	♭7	7th chord	Parent major
C	C	D	**E♭**	F	G	A	B♭	Cm7	B♭
G	G	A	**B♭**	C	D	E	F	Gm7	F
D	D	E	**F**	G	A	B	C	Dm7	C
A	A	B	**C**	D	E	F♯	G	Am7	G
E	E	F♯	**G**	A	B	C♯	D	Em7	D
B	B	C♯	**D**	E	F♯	G♯	A	Bm7	A
F♯	F♯	G♯	**A**	B	C♯	D♯	E	F♯m7	E
C♯	C♯	D♯	**E**	F♯	G♯	A♯	B	C♯m7	B
A♭	A♭	B♭	**C♭**	D♭	E♭	F	G♭	A♭m7	G♭
E♭	E♭	F	**G♭**	A♭	B♭	C	D♭	E♭m7	D♭
B♭	B♭	C	**D♭**	E♭	F	G	A♭	B♭m7	A♭
F	F	G	**A♭**	B♭	C	D	E♭	Fm7	E♭

Aeolian major with ♭6 · parent a minor 3rd up

Root	1	2	♭3	4	5	♭6	♭7	7th chord	Parent major
C	C	D	E♭	F	G	**A♭**	B♭	Cm7	E♭
G	G	A	B♭	C	D	**E♭**	F	Gm7	B♭
D	D	E	F	G	A	**B♭**	C	Dm7	F
A	A	B	C	D	E	**F**	G	Am7	C
E	E	F♯	G	A	B	**C**	D	Em7	G
B	B	C♯	D	E	F♯	**G**	A	Bm7	D
F♯	F♯	G♯	A	B	C♯	**D**	E	F♯m7	A
C♯	C♯	D♯	E	F♯	G♯	**A**	B	C♯m7	E
G♯	G♯	A♯	B	C♯	D♯	**E**	F♯	G♯m7	B
E♭	E♭	F	G♭	A♭	B♭	**C♭**	D♭	E♭m7	G♭
B♭	B♭	C	D♭	E♭	F	**G♭**	A♭	B♭m7	D♭
F	F	G	A♭	B♭	C	**D♭**	E♭	Fm7	A♭

Phrygian major with ♭2 · parent a major 3rd down

Root	1	♭2	♭3	4	5	♭6	♭7	7th chord	Parent major
C	C	**D♭**	E♭	F	G	A♭	B♭	Cm7	A♭
G	G	**A♭**	B♭	C	D	E♭	F	Gm7	E♭
D	D	**E♭**	F	G	A	B♭	C	Dm7	B♭
A	A	**B♭**	C	D	E	F	G	Am7	F
E	E	**F**	G	A	B	C	D	Em7	C
B	B	**C**	D	E	F♯	G	A	Bm7	G
F♯	F♯	**G**	A	B	C♯	D	E	F♯m7	D
C♯	C♯	**D**	E	F♯	G♯	A	B	C♯m7	A
G♯	G♯	**A**	B	C♯	D♯	E	F♯	G♯m7	E
D♯	D♯	**E**	F♯	G♯	A♯	B	C♯	D♯m7	B
B♭	B♭	**C♭**	D♭	E♭	F	G♭	A♭	B♭m7	G♭
F	F	**G♭**	A♭	B♭	C	D♭	E♭	Fm7	D♭

Locrian major with ♭5 · parent a semitone up

Root	1	♭2	♭3	4	♭5	♭6	♭7	7th chord	Parent major
C	C	D♭	E♭	F	**G♭**	A♭	B♭	Cm7♭5	D♭
G	G	A♭	B♭	C	**D♭**	E♭	F	Gm7♭5	A♭
D	D	E♭	F	G	**A♭**	B♭	C	Dm7♭5	E♭
A	A	B♭	C	D	**E♭**	F	G	Am7♭5	B♭
E	E	F	G	A	**B♭**	C	D	Em7♭5	F
B	B	C	D	E	**F**	G	A	Bm7♭5	C
F♯	F♯	G	A	B	**C**	D	E	F♯m7♭5	G
C♯	C♯	D	E	F♯	**G**	A	B	C♯m7♭5	D
G♯	G♯	A	B	C♯	**D**	E	F♯	G♯m7♭5	A
D♯	D♯	E	F♯	G♯	**A**	B	C♯	D♯m7♭5	E
A♯	A♯	B	C♯	D♯	**E**	F♯	G♯	A♯m7♭5	B
F	F	G♭	A♭	B♭	**C♭**	D♭	E♭	Fm7♭5	G♭

Contrast pairs: the note that separates them

Pair	Differs at	First	Second
Lydian vs **Ionian**	♯4 vs 4	1 2 3 ♯4 5 6 7	1 2 3 4 5 6 7
Ionian vs **Mixolydian**	7 vs ♭7	1 2 3 4 5 6 7	1 2 3 4 5 6 ♭7
Mixolydian vs **Dorian**	3 vs ♭3	1 2 3 4 5 6 ♭7	1 2 ♭3 4 5 6 ♭7
Dorian vs **Aeolian**	6 vs ♭6	1 2 ♭3 4 5 6 ♭7	1 2 ♭3 4 5 ♭6 ♭7
Aeolian vs **Phrygian**	2 vs ♭2	1 2 ♭3 4 5 ♭6 ♭7	1 ♭2 ♭3 4 5 ♭6 ♭7
Phrygian vs **Locrian**	5 vs ♭5	1 ♭2 ♭3 4 5 ♭6 ♭7	1 ♭2 ♭3 4 ♭5 ♭6 ♭7
Dorian vs **Melodic minor**	♭7 vs 7	1 2 ♭3 4 5 6 ♭7	1 2 ♭3 4 5 6 7
Aeolian vs **Harmonic minor**	♭7 vs 7	1 2 ♭3 4 5 ♭6 ♭7	1 2 ♭3 4 5 ♭6 7
Mixolydian vs **Lydian dominant**	4 vs ♯4	1 2 3 4 5 6 ♭7	1 2 3 ♯4 5 6 ♭7
Phrygian vs **Phrygian dominant**	♭3 vs 3	1 ♭2 ♭3 4 5 ♭6 ♭7	1 ♭2 3 4 5 ♭6 ♭7
Locrian vs **Locrian ♮2**	♭2 vs 2	1 ♭2 ♭3 4 ♭5 ♭6 ♭7	1 2 ♭3 4 ♭5 ♭6 ♭7
Mixolydian vs **Mixolydian ♭6**	6 vs ♭6	1 2 3 4 5 6 ♭7	1 2 3 4 5 ♭6 ♭7
Phrygian vs **Dorian ♭2**	♭6 vs 6	1 ♭2 ♭3 4 5 ♭6 ♭7	1 ♭2 ♭3 4 5 6 ♭7
Locrian vs **Altered**	4 vs ♭4	1 ♭2 ♭3 4 ♭5 ♭6 ♭7	1 ♭2 ♭3 ♭4 ♭5 ♭6 ♭7

B2 Harmonic minor modes

B2 RULE Each harmonic-minor mode is a major mode on the same root with one note raised (Ultralocrian: Locrian with ♭4 and 𝄫7). Phrygian dominant = harmonic minor a 4th above.

A harmonic minor is A natural minor with the 7th raised: A B C D E F **G♯**. Its notes are C major with G raised to G♯. So every mode of harmonic minor is a C-major mode on the same root with that one G raised, and the name says which degree moved.

The signature sound is the **augmented 2nd** between ♭6 and 7 (F to G♯). Mode 5 matters most: harmonic minor exists in practice to give a minor key a dominant V7.

Worked example. Spell B Phrygian dominant. B Phrygian is B C D E F♯ G A (parent G major, a major 3rd down). Raise the 3rd: B C **D♯** E F♯ G A.

The seven modes

#	Mode	Twin + change	Formula	7th chord	Colour	Use
1	**Harmonic minor**	Aeolian ♮7	1 2 ♭3 4 5 ♭6 7	mMaj7	♮7	Dark tonic minor
2	**Locrian ♮6**	Locrian, raise the 6th	1 ♭2 ♭3 4 ♭5 6 ♭7	m7♭5	♮13	iiø7 in minor
3	**Ionian ♯5**	Ionian, raise the 5th	1 2 3 4 ♯5 6 7	maj7♯5	♯5	Augmented colour
4	**Dorian ♯4**	Dorian, raise the 4th	1 2 ♭3 ♯4 5 6 ♭7	m7	♯11	Exotic iv
5	**Phrygian dominant**	Phrygian, raise the 3rd	1 ♭2 3 4 5 ♭6 ♭7	7	♭9, ♭13	V7♭9 into minor
6	**Lydian ♯2**	Lydian, raise the 2nd	1 ♯2 3 ♯4 5 6 7	maj7	♯9, ♯11	Exotic ♭VI
7	**Ultralocrian**	Locrian, lower the 4th and 7th	1 ♭2 ♭3 ♭4 ♭5 ♭6 𝄫7	°7	dim7	vii°7 into i

Every harmonic-minor mode on all 12 roots

Highlighted: the note(s) that differ from the twin major mode. Last column: the harmonic minor that contains it.

Harmonic minor Aeolian, raise the 7th

Root	1	2	♭3	4	5	♭6	7	7th chord	Parent harm. min.
C	C	D	E♭	F	G	A♭	**B**	CmMaj7	C
G	G	A	B♭	C	D	E♭	**F♯**	GmMaj7	G
D	D	E	F	G	A	B♭	**C♯**	DmMaj7	D
A	A	B	C	D	E	F	**G♯**	AmMaj7	A
E	E	F♯	G	A	B	C	**D♯**	EmMaj7	E
B	B	C♯	D	E	F♯	G	**A♯**	BmMaj7	B
F♯	F♯	G♯	A	B	C♯	D	**E♯**	F♯mMaj7	F♯
C♯	C♯	D♯	E	F♯	G♯	A	**B♯**	C♯mMaj7	C♯
A♭	A♭	B♭	C♭	D♭	E♭	F♭	**G**	A♭mMaj7	A♭
E♭	E♭	F	G♭	A♭	B♭	C♭	**D**	E♭mMaj7	E♭
B♭	B♭	C	D♭	E♭	F	G♭	**A**	B♭mMaj7	B♭
F	F	G	A♭	B♭	C	D♭	**E**	FmMaj7	F

Locrian ♮6 Locrian, raise the 6th

Root	1	♭2	♭3	4	♭5	6	♭7	7th chord	Parent harm. min.
C	C	D♭	E♭	F	G♭	**A**	B♭	Cm7♭5	B♭
G	G	A♭	B♭	C	D♭	**E**	F	Gm7♭5	F
D	D	E♭	F	G	A♭	**B**	C	Dm7♭5	C
A	A	B♭	C	D	E♭	**F♯**	G	Am7♭5	G
E	E	F	G	A	B♭	**C♯**	D	Em7♭5	D
B	B	C	D	E	F	**G♯**	A	Bm7♭5	A
F♯	F♯	G	A	B	C	**D♯**	E	F♯m7♭5	E
C♯	C♯	D	E	F♯	G	**A♯**	B	C♯m7♭5	B
G♯	G♯	A	B	C♯	D	**E♯**	F♯	G♯m7♭5	F♯
D♯	D♯	E	F♯	G♯	A	**B♯**	C♯	D♯m7♭5	C♯
B♭	B♭	C♭	D♭	E♭	F♭	**G**	A♭	B♭m7♭5	A♭
F	F	G♭	A♭	B♭	C♭	**D**	E♭	Fm7♭5	E♭

Ionian ♯5 Ionian, raise the 5th

Root	1	2	3	4	♯5	6	7	7th chord	Parent harm. min.
C	C	D	E	F	**G♯**	A	B	Cmaj7♯5	A
G	G	A	B	C	**D♯**	E	F♯	Gmaj7♯5	E
D	D	E	F♯	G	**A♯**	B	C♯	Dmaj7♯5	B
A	A	B	C♯	D	**E♯**	F♯	G♯	Amaj7♯5	F♯
E	E	F♯	G♯	A	**B♯**	C♯	D♯	Emaj7♯5	C♯
B	B	C♯	D♯	E	**F𝄪**	G♯	A♯	Bmaj7♯5	G♯
G♭	G♭	A♭	B♭	C♭	**D**	E♭	F	G♭maj7♯5	E♭
D♭	D♭	E♭	F	G♭	**A**	B♭	C	D♭maj7♯5	B♭
A♭	A♭	B♭	C	D♭	**E**	F	G	A♭maj7♯5	F
E♭	E♭	F	G	A♭	**B**	C	D	E♭maj7♯5	C
B♭	B♭	C	D	E♭	**F♯**	G	A	B♭maj7♯5	G
F	F	G	A	B♭	**C♯**	D	E	Fmaj7♯5	D

Dorian ♯4 Dorian, raise the 4th

Root	1	2	♭3	♯4	5	6	♭7	7th chord	Parent harm. min.
C	C	D	E♭	**F♯**	G	A	B♭	Cm7	G
G	G	A	B♭	**C♯**	D	E	F	Gm7	D
D	D	E	F	**G♯**	A	B	C	Dm7	A
A	A	B	C	**D♯**	E	F♯	G	Am7	E
E	E	F♯	G	**A♯**	B	C♯	D	Em7	B
B	B	C♯	D	**E♯**	F♯	G♯	A	Bm7	F♯
F♯	F♯	G♯	A	**B♯**	C♯	D♯	E	F♯m7	C♯
D♭	D♭	E♭	F♭	**G**	A♭	B♭	C♭	D♭m7	A♭
A♭	A♭	B♭	C♭	**D**	E♭	F	G♭	A♭m7	E♭
E♭	E♭	F	G♭	**A**	B♭	C	D♭	E♭m7	B♭
B♭	B♭	C	D♭	**E**	F	G	A♭	B♭m7	F
F	F	G	A♭	**B**	C	D	E♭	Fm7	C

Phrygian dominant Phrygian, raise the 3rd

Root	1	♭2	3	4	5	♭6	♭7	7th chord	Parent harm. min.
C	C	D♭	**E**	F	G	A♭	B♭	C7	F
G	G	A♭	**B**	C	D	E♭	F	G7	C
D	D	E♭	**F♯**	G	A	B♭	C	D7	G
A	A	B♭	**C♯**	D	E	F	G	A7	D
E	E	F	**G♯**	A	B	C	D	E7	A
B	B	C	**D♯**	E	F♯	G	A	B7	E
F♯	F♯	G	**A♯**	B	C♯	D	E	F♯7	B
C♯	C♯	D	**E♯**	F♯	G♯	A	B	C♯7	F♯
G♯	G♯	A	**B♯**	C♯	D♯	E	F♯	G♯7	C♯
E♭	E♭	F♭	**G**	A♭	B♭	C♭	D♭	E♭7	A♭
B♭	B♭	C♭	**D**	E♭	F	G♭	A♭	B♭7	E♭
F	F	G♭	**A**	B♭	C	D♭	E♭	F7	B♭

Lydian ♯2 Lydian, raise the 2nd

Root	1	♯2	3	♯4	5	6	7	7th chord	Parent harm. min.
C	C	**D♯**	E	F♯	G	A	B	Cmaj7	E
G	G	**A♯**	B	C♯	D	E	F♯	Gmaj7	B
D	D	**E♯**	F♯	G♯	A	B	C♯	Dmaj7	F♯
A	A	**B♯**	C♯	D♯	E	F♯	G♯	Amaj7	C♯
E	E	**F𝄪**	G♯	A♯	B	C♯	D♯	Emaj7	G♯
B	B	**C𝄪**	D♯	E♯	F♯	G♯	A♯	Bmaj7	D♯
G♭	G♭	**A**	B♭	C	D♭	E♭	F	G♭maj7	B♭
D♭	D♭	**E**	F	G	A♭	B♭	C	D♭maj7	F
A♭	A♭	**B**	C	D	E♭	F	G	A♭maj7	C
E♭	E♭	**F♯**	G	A	B♭	C	D	E♭maj7	G
B♭	B♭	**C♯**	D	E	F	G	A	B♭maj7	D
F	F	**G♯**	A	B	C	D	E	Fmaj7	A

Ultralocrian Locrian, lower the 4th and 7th

Root	1	♭2	♭3	♭4	♭5	♭6	𝄫7	7th chord	Parent harm. min.
C	C	D♭	E♭	**F♭**	G♭	A♭	**B𝄫**	C°7	D♭
G	G	A♭	B♭	**C♭**	D♭	E♭	**F♭**	G°7	A♭
D	D	E♭	F	**G♭**	A♭	B♭	**C♭**	D°7	E♭
A	A	B♭	C	**D♭**	E♭	F	**G♭**	A°7	B♭
E	E	F	G	**A♭**	B♭	C	**D♭**	E°7	F
B	B	C	D	**E♭**	F	G	**A♭**	B°7	C
F♯	F♯	G	A	**B♭**	C	D	**E♭**	F♯°7	G
C♯	C♯	D	E	**F**	G	A	**B♭**	C♯°7	D
G♯	G♯	A	B	**C**	D	E	**F**	G♯°7	A
D♯	D♯	E	F♯	**G**	A	B	**C**	D♯°7	E
A♯	A♯	B	C♯	**D**	E	F♯	**G**	A♯°7	B
F	F	G♭	A♭	**B𝄫**	C♭	D♭	**E𝄫**	F°7	G♭

B3 Melodic minor modes

B3 RULE Each melodic-minor mode is a major mode with one note changed. X7alt = melodic minor a semitone above X; X7♯11 = a 5th above; Xm7♭5 = a minor 3rd above.

A melodic minor (jazz form) is A B C D E F♯ G♯: A major with a ♭3. Its notes are G major with G raised to G♯. So every mode is a G-major mode on the same root with that one G raised.

On the **Altered** scale, read the degrees as a dominant chord: ♭2 is ♭9, ♭3 is ♯9, ♭4 is the major 3rd, ♭5 is ♯11, ♭6 is ♭13.

Parent shortcuts:

- X7alt: melodic minor a **semitone above** X. G7alt = A♭ melodic minor.
- X7♯11 (Lydian dominant): melodic minor a **5th above** X. D7♯11 = A melodic minor.
- Xm7♭5 (Locrian ♮2): melodic minor a **minor 3rd above** X. F♯m7♭5 = A melodic minor.
- X Phrygian dominant: harmonic minor a **4th above** X. E Phrygian dominant = A harmonic minor.

Worked example. Spell D♭ Lydian dominant. D♭ Mixolydian is D♭ E♭ F G♭ A♭ B♭ C♭ (parent G♭ major). Raise the 4th: D♭ E♭ F **G** A♭ B♭ C♭. Check: a 5th above D♭ is A♭, and A♭ melodic minor has the same notes.

The seven modes

#	Mode	Twin + change	Formula	7th chord	Colour	Use
1	**Melodic minor**	Dorian ♮7 (or Ionian ♭3)	1 2 ♭3 4 5 6 7	mMaj7	6, 9	Tonic minor (m6/9)
2	**Dorian ♭2**	Phrygian, raise the 6th	1 ♭2 ♭3 4 5 6 ♭7	m7	♭9	7sus♭9 sound
3	**Lydian augmented**	Lydian, raise the 5th	1 2 3 ♯4 ♯5 6 7	maj7♯5	♯11	Floating maj7♯5
4	**Lydian dominant**	Mixolydian, raise the 4th	1 2 3 ♯4 5 6 ♭7	7	♯11	IV7, ♭VII7, tritone subs
5	**Mixolydian ♭6**	Aeolian, raise the 3rd	1 2 3 4 5 ♭6 ♭7	7	♭13	Gentle V7 in minor
6	**Locrian ♮2**	Locrian, raise the 2nd	1 2 ♭3 4 ♭5 ♭6 ♭7	m7♭5	♮9	iiø7 in minor (standard)
7	**Altered**	Locrian, lower the 4th	1 ♭2 ♭3 ♭4 ♭5 ♭6 ♭7	7alt	♭9, ♯9, ♯11, ♭13	Tense V7 into major or minor

Every melodic-minor mode on all 12 roots

Highlighted: the note that differs from the twin major mode. Altered is spelled as a scale (♭4); over the chord read the ♭4 as the 3rd.

Melodic minor Dorian, raise the 7th

Root	1	2	♭3	4	5	6	7	7th chord	Parent mel. min.
C	C	D	E♭	F	G	A	**B**	CmMaj7	C
G	G	A	B♭	C	D	E	**F♯**	GmMaj7	G
D	D	E	F	G	A	B	**C♯**	DmMaj7	D
A	A	B	C	D	E	F♯	**G♯**	AmMaj7	A
E	E	F♯	G	A	B	C♯	**D♯**	EmMaj7	E
B	B	C♯	D	E	F♯	G♯	**A♯**	BmMaj7	B
F♯	F♯	G♯	A	B	C♯	D♯	**E♯**	F♯mMaj7	F♯
C♯	C♯	D♯	E	F♯	G♯	A♯	**B♯**	C♯mMaj7	C♯
A♭	A♭	B♭	C♭	D♭	E♭	F	**G**	A♭mMaj7	A♭
E♭	E♭	F	G♭	A♭	B♭	C	**D**	E♭mMaj7	E♭
B♭	B♭	C	D♭	E♭	F	G	**A**	B♭mMaj7	B♭
F	F	G	A♭	B♭	C	D	**E**	FmMaj7	F

Dorian ♭2 Phrygian, raise the 6th

Root	1	♭2	♭3	4	5	6	♭7	7th chord	Parent mel. min.
C	C	D♭	E♭	F	G	**A**	B♭	Cm7	B♭
G	G	A♭	B♭	C	D	**E**	F	Gm7	F
D	D	E♭	F	G	A	**B**	C	Dm7	C
A	A	B♭	C	D	E	**F♯**	G	Am7	G
E	E	F	G	A	B	**C♯**	D	Em7	D
B	B	C	D	E	F♯	**G♯**	A	Bm7	A
F♯	F♯	G	A	B	C♯	**D♯**	E	F♯m7	E
C♯	C♯	D	E	F♯	G♯	**A♯**	B	C♯m7	B
G♯	G♯	A	B	C♯	D♯	**E♯**	F♯	G♯m7	F♯
E♭	E♭	F♭	G♭	A♭	B♭	**C**	D♭	E♭m7	D♭
B♭	B♭	C♭	D♭	E♭	F	**G**	A♭	B♭m7	A♭
F	F	G♭	A♭	B♭	C	**D**	E♭	Fm7	E♭

Lydian augmented Lydian, raise the 5th

Root	1	2	3	♯4	♯5	6	7	7th chord	Parent mel. min.
C	C	D	E	F♯	**G♯**	A	B	Cmaj7♯5	A
G	G	A	B	C♯	**D♯**	E	F♯	Gmaj7♯5	E
D	D	E	F♯	G♯	**A♯**	B	C♯	Dmaj7♯5	B
A	A	B	C♯	D♯	**E♯**	F♯	G♯	Amaj7♯5	F♯
E	E	F♯	G♯	A♯	**B♯**	C♯	D♯	Emaj7♯5	C♯
B	B	C♯	D♯	E♯	**F𝄪**	G♯	A♯	Bmaj7♯5	G♯
G♭	G♭	A♭	B♭	C	**D**	E♭	F	G♭maj7♯5	E♭
D♭	D♭	E♭	F	G	**A**	B♭	C	D♭maj7♯5	B♭
A♭	A♭	B♭	C	D	**E**	F	G	A♭maj7♯5	F
E♭	E♭	F	G	A	**B**	C	D	E♭maj7♯5	C
B♭	B♭	C	D	E	**F♯**	G	A	B♭maj7♯5	G
F	F	G	A	B	**C♯**	D	E	Fmaj7♯5	D

Lydian dominant Mixolydian, raise the 4th

Root	1	2	3	♯4	5	6	♭7	7th chord	Parent mel. min.
C	C	D	E	**F♯**	G	A	B♭	C7	G
G	G	A	B	**C♯**	D	E	F	G7	D
D	D	E	F♯	**G♯**	A	B	C	D7	A
A	A	B	C♯	**D♯**	E	F♯	G	A7	E
E	E	F♯	G♯	**A♯**	B	C♯	D	E7	B
B	B	C♯	D♯	**E♯**	F♯	G♯	A	B7	F♯
F♯	F♯	G♯	A♯	**B♯**	C♯	D♯	E	F♯7	C♯
D♭	D♭	E♭	F	**G**	A♭	B♭	C♭	D♭7	A♭
A♭	A♭	B♭	C	**D**	E♭	F	G♭	A♭7	E♭
E♭	E♭	F	G	**A**	B♭	C	D♭	E♭7	B♭
B♭	B♭	C	D	**E**	F	G	A♭	B♭7	F
F	F	G	A	**B**	C	D	E♭	F7	C

Mixolydian ♭6 Aeolian, raise the 3rd

Root	1	2	3	4	5	♭6	♭7	7th chord	Parent mel. min.
C	C	D	**E**	F	G	A♭	B♭	C7	F
G	G	A	**B**	C	D	E♭	F	G7	C
D	D	E	**F♯**	G	A	B♭	C	D7	G
A	A	B	**C♯**	D	E	F	G	A7	D
E	E	F♯	**G♯**	A	B	C	D	E7	A
B	B	C♯	**D♯**	E	F♯	G	A	B7	E
F♯	F♯	G♯	**A♯**	B	C♯	D	E	F♯7	B
C♯	C♯	D♯	**E♯**	F♯	G♯	A	B	C♯7	F♯
A♭	A♭	B♭	**C**	D♭	E♭	F♭	G♭	A♭7	D♭
E♭	E♭	F	**G**	A♭	B♭	C♭	D♭	E♭7	A♭
B♭	B♭	C	**D**	E♭	F	G♭	A♭	B♭7	E♭
F	F	G	**A**	B♭	C	D♭	E♭	F7	B♭

Locrian ♮2 Locrian, raise the 2nd

Root	1	2	♭3	4	♭5	♭6	♭7	7th chord	Parent mel. min.
C	C	**D**	E♭	F	G♭	A♭	B♭	Cm7♭5	E♭
G	G	**A**	B♭	C	D♭	E♭	F	Gm7♭5	B♭
D	D	**E**	F	G	A♭	B♭	C	Dm7♭5	F
A	A	**B**	C	D	E♭	F	G	Am7♭5	C
E	E	**F♯**	G	A	B♭	C	D	Em7♭5	G
B	B	**C♯**	D	E	F	G	A	Bm7♭5	D
F♯	F♯	**G♯**	A	B	C	D	E	F♯m7♭5	A
C♯	C♯	**D♯**	E	F♯	G	A	B	C♯m7♭5	E
G♯	G♯	**A♯**	B	C♯	D	E	F♯	G♯m7♭5	B
D♯	D♯	**E♯**	F♯	G♯	A	B	C♯	D♯m7♭5	F♯
B♭	B♭	**C**	D♭	E♭	F♭	G♭	A♭	B♭m7♭5	D♭
F	F	**G**	A♭	B♭	C♭	D♭	E♭	Fm7♭5	A♭

Altered Locrian, lower the 4th

Root	1	♭2	♭3	♭4	♭5	♭6	♭7	7th chord	Parent mel. min.
C	C	D♭	E♭	**F♭**	G♭	A♭	B♭	C7alt	D♭
G	G	A♭	B♭	**C♭**	D♭	E♭	F	G7alt	A♭
D	D	E♭	F	**G♭**	A♭	B♭	C	D7alt	E♭
A	A	B♭	C	**D♭**	E♭	F	G	A7alt	B♭
E	E	F	G	**A♭**	B♭	C	D	E7alt	F
B	B	C	D	**E♭**	F	G	A	B7alt	C
F♯	F♯	G	A	**B♭**	C	D	E	F♯7alt	G
C♯	C♯	D	E	**F**	G	A	B	C♯7alt	D
G♯	G♯	A	B	**C**	D	E	F♯	G♯7alt	A
D♯	D♯	E	F♯	**G**	A	B	C♯	D♯7alt	E
A♯	A♯	B	C♯	**D**	E	F♯	G♯	A♯7alt	B
F	F	G♭	A♭	**B𝄫**	C♭	D♭	E♭	F7alt	G♭

B4 Symmetric scales

B4 RULE Half-whole: H W repeating, over 7♭9 (repeats every minor 3rd). Whole-half: W H, over dim7. Whole tone: all whole steps, over 7♯5 (no perfect 5th).

These scales repeat a fixed step pattern, so on the pitch clock they make regular shapes and there are only a few of each.

The half-whole scale has ♭9, ♯9, ♯11 and ♮**13** with a **natural 5th**, unlike the altered scale, which has ♭13 and no natural 5th. That is the contrast to drill.

Scale	Steps	Formula	Over	Versions	Shortcut
Half-whole diminished	H W H W H W H W	1 ♭2 ♯2 3 ♯4 5 6 ♭7	7♭9, 13♭9	3	Repeats every minor 3rd: G7, B♭7, D♭7, E7 share one scale
Whole-half diminished	W H W H W H W H	1 2 ♭3 4 ♭5 ♭6 6 7	dim7	3	Same notes as half-whole a semitone lower
Whole tone	W W W W W W	1 2 3 ♯4 ♯5 ♭7	7♯5, 9♯5	2	Repeats every whole step; contains no perfect 5th

Half-whole diminished: 3 sets of 4 dominants

Set	Root	Notes	Fits
1	**C**	C D♭ D♯ E F♯ G A B♭	C7♭9
	E♭	E♭ F♭ F♯ G A B♭ C D♭	E♭7♭9
	F♯	F♯ G G𝄪 A♯ B♯ C♯ D♯ E	F♯7♭9
	A	A B♭ B♯ C♯ D♯ E F♯ G	A7♭9
2	**D♭**	D♭ E𝄫 E F G A♭ B♭ C♭	D♭7♭9
	E	E F F𝄪 G♯ A♯ B C♯ D	E7♭9
	G	G A♭ A♯ B C♯ D E F	G7♭9
	B♭	B♭ C♭ C♯ D E F G A♭	B♭7♭9
3	**D**	D E♭ E♯ F♯ G♯ A B C	D7♭9
	F	F G♭ G♯ A B C D E♭	F7♭9
	A♭	A♭ B𝄫 B C D E♭ F G♭	A♭7♭9
	B	B C C𝄪 D♯ E♯ F♯ G♯ A	B7♭9

Whole-half diminished: 3 sets of 4 dim7 chords

Set	Root	Notes	Fits
1	**C**	C D E♭ F G♭ A♭ A B	C°7
	E♭	E♭ F G♭ A♭ B𝄫 C♭ C D	E♭°7
	F♯	F♯ G♯ A B C D D♯ E♯	F♯°7
	A	A B C D E♭ F F♯ G♯	A°7
2	**C♯**	C♯ D♯ E F♯ G A A♯ B♯	C♯°7
	E	E F♯ G A B♭ C C♯ D♯	E°7
	G	G A B♭ C D♭ E♭ E F♯	G°7
	B♭	B♭ C D♭ E♭ F♭ G♭ G A	B♭°7
3	**D**	D E F G A♭ B♭ B C♯	D°7
	F	F G A♭ B♭ C♭ D♭ D E	F°7
	G♯	G♯ A♯ B C♯ D E E♯ F𝄪	G♯°7
	B	B C♯ D E F G G♯ A♯	B°7

Whole tone: 2 sets of 6

Set	Root	Notes	Fits
1	**C**	C D E F♯ G♯ B♭	C7♯5
	D	D E F♯ G♯ A♯ C	D7♯5
	E	E F♯ G♯ A♯ B♯ D	E7♯5
	G♭	G♭ A♭ B♭ C D F♭	G♭7♯5
	A♭	A♭ B♭ C D E G♭	A♭7♯5
	B♭	B♭ C D E F♯ A♭	B♭7♯5
2	**D♭**	D♭ E♭ F G A C♭	D♭7♯5
	E♭	E♭ F G A B D♭	E♭7♯5
	F	F G A B C♯ E♭	F7♯5
	G	G A B C♯ D♯ F	G7♯5
	A	A B C♯ D♯ E♯ G	A7♯5
	B	B C♯ D♯ E♯ F𝄪 A	B7♯5

Set membership is the memory fact. Half-whole set 1 = C, E♭, F♯, A dominants; Half-whole set 2 = D♭, E, G, B♭ dominants; Half-whole set 3 = D, F, A♭, B dominants. Whole-half on a note = half-whole a semitone lower. Whole-tone set 1 = C D E F♯ G♯ B♭; set 2 = D♭ E♭ F G A C♭.

B5 Parent shortcuts

The fastest way to spell a melodic- or harmonic-minor mode is to jump to its parent scale and read it from your root. These are the jumps worth knowing cold; the mode tables in B1–B3 show every root.

Chord	Mode	Parent jump	Example	Read from the root
X7alt	Altered	mel. minor ½ step up	G7alt → A♭ mel. minor: A♭ B♭ C♭ D♭ E♭ F G	G Altered: G A♭ B♭ C♭ D♭ E♭ F
X7♯11	Lydian dominant	mel. minor 5th up	D7 → A mel. minor: A B C D E F♯ G♯	D Lydian dominant: D E F♯ G♯ A B C
Xm7♭5	Locrian ♮2	mel. minor m3 up	F♯m7♭5 → A mel. minor: A B C D E F♯ G♯	F♯ Locrian ♮2: F♯ G♯ A B C D E
X7♭9 (to minor)	Phrygian dominant	harm. minor 4th up	E7 → A harm. minor: A B C D E F G♯	E Phrygian dominant: E F G♯ A B C D
X7♭13	Mixolydian ♭6	mel. minor 4th up	E7 → A mel. minor: A B C D E F♯ G♯	E Mixolydian ♭6: E F♯ G♯ A B C D
Xm7 (Dorian)	Dorian	major a whole step down	Dm7 → C major: C D E F G A B	D Dorian: D E F G A B C

The four to know cold: X7alt → melodic minor a **semitone up**. X7♯11 → melodic minor a **5th up**. Xm7♭5 → melodic minor a **minor 3rd up**. X Phrygian dominant → harmonic minor a **4th up**.

One scale, four chords: each melodic minor covers its own m6/mMaj7, the m7♭5 a major 6th up, the 7♯11 a 4th up and the 7alt a major 7th up. A♭ melodic minor = A♭mMaj7 = Fm7♭5 = D♭7♯11 = G7alt.

Major-mode parents (from B1)

Mode	Parent	Example
Lydian	a 4th down	F Lydian = C major from F
Mixolydian	a 5th down	F Mixolydian = B♭ major from F
Dorian	a whole step down	F Dorian = E♭ major from F
Aeolian	a minor 3rd up	F Aeolian = A♭ major from F
Phrygian	a major 3rd down	F Phrygian = D♭ major from F
Locrian	a semitone up	F Locrian = G♭ major from F

B6 Chord → scale cheat sheet

First choice = the scale that names the chord's own colour without an avoid note. When a chord has more than one job, the job decides the scale.

Chord	Typical job	First choice	Also good	Remember
maj7, maj9, 6, 6/9	I (tonic major)	**Ionian**	Major pentatonic; Lydian for a brighter tonic	The 4th (11) is the avoid note over maj7
maj7♯11, maj7 on IV	IV, ♭VI, ♭II, ♭III	**Lydian**	II triad / II major pentatonic on top	Parent: major a 4th down
maj7♯5	Augmented colour	**Lydian augmented**	Ionian ♯5 (harmonic minor mode 3)	Parent: melodic minor a minor 3rd down
m7, m9, m11	ii	**Dorian**	ii minor pentatonic (Dorian pentatonic)	♮13 is the Dorian colour
m7	vi, or i in natural minor	**Aeolian**	Dorian for a brighter minor	♭13 is the Aeolian colour
m7	iii	**Phrygian**	Aeolian if the melody has a ♮9	♭9 is a colour note, not an avoid note in modal use
m6, m6/9, mMaj7	Tonic minor (jazz)	**Melodic minor**	Dorian	V triad on top gives m(maj9)
mMaj7 (dark)	Tonic minor (classical)	**Harmonic minor**	Melodic minor	The ♭6–7 augmented 2nd is the sound
m7♭5	iiø7 in minor	**Locrian ♮2**	Locrian; Locrian ♮6	Locrian ♮2 = mel. minor a m3 up
°7	vii°7, passing diminished	**Whole-half diminished**	Ultralocrian (inside a harmonic-minor key)	Any note of a dim7 can be the root
7, 9, 13	V (unaltered), blues	**Mixolydian**	v minor pentatonic for 9sus4	11 is the avoid note unless sus
7sus4, 9sus4	Suspended V	**Mixolydian**	Dorian ♭2 for 7sus♭9	The 4th replaces the 3rd
7♯11, 9♯11, 13♯11	IV7, ♭VII7, subV7, II7	**Lydian dominant**	II triad on top	Mel. minor a 5th up
7♭9, 7♭9♭13	V into a minor chord	**Phrygian dominant**	Altered	Harm. minor a 4th up (the key you resolve to)
7♭13, 9♭13	Gentle V in minor	**Mixolydian ♭6**	Phrygian dominant	Mel. minor a 4th up
13♭9, 7♭9 (♮13)	V into major, bright tension	**Half-whole diminished**	VI triad on top	Has ♮5 and ♮13; repeats every minor 3rd
7alt, 7♯9♭13, 7♭9♯9	V into major or minor, max tension	**Altered**	♭iii minor pentatonic; ♭VI or ♭V triads	Mel. minor a semitone up; no ♮5
7♯5, 9♯5	Augmented dominant	**Whole tone**	Altered	No perfect 5th; repeats every whole step

Dominant decision tree: resolving to major and want it inside → Mixolydian. Bright tension → half-whole (13♭9). Maximum tension → altered. Resolving to minor → Phrygian dominant (♭9 ♭13) or altered. Not resolving down a 5th (IV7, ♭VII7, subV) → Lydian dominant.

C1 Secondary dominants

C1 RULE Go up a 5th from the target and make it a dominant 7th. Its 3rd is the target's leading tone (V7/IV adds the ♭7 instead).

A V7 pulls to I because its 3rd is a semitone below the tonic and its 3rd and 7th form a tritone that wants to resolve. You can point that pull at any diatonic chord (except the diminished vii) by playing that chord's own V7 first. **The rule: go up a 5th from the target and make it a dominant 7th.** Each secondary dominant adds one note from outside the key, its 3rd, which is the leading tone of the target. (V7/IV adds the ♭7 instead.)

When the target is minor (ii, iii, vi), its ii–V is a minor ii–V: m7♭5 then 7♭9 or 7alt.

In a minor key the same rule gives, in A minor: V7/iv = A7, V7/v = B7, V7/♭VI = C7, V7/♭VII = D7. (V7/♭III is G7, already diatonic as ♭VII7.)

Secondary dominants in all 12 major keys

Each cell: the secondary dominant, its characteristic out-of-key note (accent: the leading tone of the target, or the ♭7 for V7/IV), any other out-of-key note (grey), and the ii–V into the target.

Key	V7/ii (VI7)	V7/iii (VII7)	V7/IV (I7)	V7/V (II7)	V7/vi (III7)
C	**A7 C♯** Em7♭5–A7–Dm7	**B7 D♯** +F♯ F♯m7♭5–B7–Em7	**C7 B♭** Gm7–C7–Fmaj7	**D7 F♯** Am7–D7–G7	**E7 G♯** Bm7♭5–E7–Am7
G	**E7 G♯** Bm7♭5–E7–Am7	**F♯7 A♯** +C♯ C♯m7♭5–F♯7–Bm7	**G7 F** Dm7–G7–Cmaj7	**A7 C♯** Em7–A7–D7	**B7 D♯** F♯m7♭5–B7–Em7
D	**B7 D♯** F♯m7♭5–B7–Em7	**C♯7 E♯** +G♯ G♯m7♭5–C♯7–F♯m7	**D7 C** Am7–D7–Gmaj7	**E7 G♯** Bm7–E7–A7	**F♯7 A♯** C♯m7♭5–F♯7–Bm7
A	**F♯7 A♯** C♯m7♭5–F♯7–Bm7	**G♯7 B♯** +D♯ D♯m7♭5–G♯7–C♯m7	**A7 G** Em7–A7–Dmaj7	**B7 D♯** F♯m7–B7–E7	**C♯7 E♯** G♯m7♭5–C♯7–F♯m7
E	**C♯7 E♯** G♯m7♭5–C♯7–F♯m7	**D♯7 F𝄪** +A♯ A♯m7♭5–D♯7–G♯m7	**E7 D** Bm7–E7–Amaj7	**F♯7 A♯** C♯m7–F♯7–B7	**G♯7 B♯** D♯m7♭5–G♯7–C♯m7
B	**G♯7 B♯** D♯m7♭5–G♯7–C♯m7	**A♯7 C𝄪** +E♯ E♯m7♭5–A♯7–D♯m7	**B7 A** F♯m7–B7–Emaj7	**C♯7 E♯** G♯m7–C♯7–F♯7	**D♯7 F𝄪** A♯m7♭5–D♯7–G♯m7
F♯	**D♯7 F𝄪** A♯m7♭5–D♯7–G♯m7	**E♯7 G𝄪** +B♯ B♯m7♭5–E♯7–A♯m7	**F♯7 E** C♯m7–F♯7–Bmaj7	**G♯7 B♯** D♯m7–G♯7–C♯7	**A♯7 C𝄪** E♯m7♭5–A♯7–D♯m7
D♭	**B♭7 D** Fm7♭5–B♭7–E♭m7	**C7 E** +G Gm7♭5–C7–Fm7	**D♭7 C♭** A♭m7–D♭7–G♭maj7	**E♭7 G** B♭m7–E♭7–A♭7	**F7 A** Cm7♭5–F7–B♭m7
A♭	**F7 A** Cm7♭5–F7–B♭m7	**G7 B** +D Dm7♭5–G7–Cm7	**A♭7 G♭** E♭m7–A♭7–D♭maj7	**B♭7 D** Fm7–B♭7–E♭7	**C7 E** Gm7♭5–C7–Fm7
E♭	**C7 E** Gm7♭5–C7–Fm7	**D7 F♯** +A Am7♭5–D7–Gm7	**E♭7 D♭** B♭m7–E♭7–A♭maj7	**F7 A** Cm7–F7–B♭7	**G7 B** Dm7♭5–G7–Cm7
B♭	**G7 B** Dm7♭5–G7–Cm7	**A7 C♯** +E Em7♭5–A7–Dm7	**B♭7 A♭** Fm7–B♭7–E♭maj7	**C7 E** Gm7–C7–F7	**D7 F♯** Am7♭5–D7–Gm7
F	**D7 F♯** Am7♭5–D7–Gm7	**E7 G♯** +B Bm7♭5–E7–Am7	**F7 E♭** Cm7–F7–B♭maj7	**G7 B** Dm7–G7–C7	**A7 C♯** Em7♭5–A7–Dm7

Minor keys

Key	V7/iv	V7/v	V7/♭VI	V7/♭VII	V7 (harmonic minor)
Am	**A7 C♯**	**B7 D♯ F♯**	**C7 B♭**	**D7 F♯**	**E7 G♯**
Em	**E7 G♯**	**F♯7 A♯ C♯**	**G7 F**	**A7 C♯**	**B7 D♯**
Bm	**B7 D♯**	**C♯7 E♯ G♯**	**D7 C**	**E7 G♯**	**F♯7 A♯**
F♯m	**F♯7 A♯**	**G♯7 B♯ D♯**	**A7 G**	**B7 D♯**	**C♯7 E♯**
C♯m	**C♯7 E♯**	**D♯7 F𝄪 A♯**	**E7 D**	**F♯7 A♯**	**G♯7 B♯**
G♯m	**G♯7 B♯**	**A♯7 C𝄪 E♯**	**B7 A**	**C♯7 E♯**	**D♯7 F𝄪**
D♯m	**D♯7 F𝄪**	**E♯7 G𝄪 B♯**	**F♯7 E**	**G♯7 B♯**	**A♯7 C𝄪**
B♭m	**B♭7 D**	**C7 E G**	**D♭7 C♭**	**E♭7 G**	**F7 A**
Fm	**F7 A**	**G7 B D**	**A♭7 G♭**	**B♭7 D**	**C7 E**
Cm	**C7 E**	**D7 F♯ A**	**E♭7 D♭**	**F7 A**	**G7 B**
Gm	**G7 B**	**A7 C♯ E**	**B♭7 A♭**	**C7 E**	**D7 F♯**
Dm	**D7 F♯**	**E7 G♯ B**	**F7 E♭**	**G7 B**	**A7 C♯**

C2 Tritone substitution

C2 RULE The dominant a tritone away has the same 3rd and 7th, swapped, and resolves down a semitone.

A dominant 7th's 3rd and 7th sit a tritone apart. The dominant whose root is a tritone away has **the same two notes, swapped**: G7 has B and F, D♭7 has F and C♭. With the same guide tones it does the same job, and its root resolves **down a semitone** instead of down a 5th, giving a chromatic bass (Dm7–D♭7–C). On the pitch clock the two roots sit directly opposite each other.

The six pairs: **C7↔G♭7, G7↔D♭7, D7↔A♭7, A7↔E♭7, E7↔B♭7, B7↔F7.**

Secondary dominants can be substituted too: in C, A7 (V7/ii) becomes E♭7, giving Cmaj7–E♭7–Dm7–D♭7–Cmaj7.

The six tritone pairs

Dominant	Guide tones (3, ♭7)	Tritone sub	Guide tones (3, ♭7)	Both resolve to
C7	E, B♭	**G♭7**	B♭, F♭	F (down a 5th / down a semitone)
G7	B, F	**D♭7**	F, C♭	C (down a 5th / down a semitone)
D7	F♯, C	**A♭7**	C, G♭	G (down a 5th / down a semitone)
A7	C♯, G	**E♭7**	G, D♭	D (down a 5th / down a semitone)
E7	G♯, D	**B♭7**	D, A♭	A (down a 5th / down a semitone)
B7	D♯, A	**F7**	A, E♭	E (down a 5th / down a semitone)

subV7 of every target in all 12 major keys

The tritone sub of V7/x is the dominant a semitone above x. Last column: subV7 of I (♭II7), the substitute for the main V7.

Key	subV7/ii	subV7/iii	subV7/IV	subV7/V	subV7/vi	subV7/I
C	**E♭7** for A7	**F7** for B7	**G♭7** for C7	**A♭7** for D7	**B♭7** for E7	**D♭7** for G7
G	**B♭7** for E7	**C7** for F♯7	**D♭7** for G7	**E♭7** for A7	**F7** for B7	**A♭7** for D7
D	**F7** for B7	**G7** for C♯7	**A♭7** for D7	**B♭7** for E7	**C7** for F♯7	**E♭7** for A7
A	**C7** for F♯7	**D7** for G♯7	**E♭7** for A7	**F7** for B7	**G7** for C♯7	**B♭7** for E7
E	**G7** for C♯7	**A7** for D♯7	**B♭7** for E7	**C7** for F♯7	**D7** for G♯7	**F7** for B7
B	**D7** for G♯7	**E7** for A♯7	**F7** for B7	**G7** for C♯7	**A7** for D♯7	**C7** for F♯7
F♯	**A7** for D♯7	**B7** for E♯7	**C7** for F♯7	**D7** for G♯7	**E7** for A♯7	**G7** for C♯7
D♭	**F♭7** for B♭7	**G♭7** for C7	**A𝄫7** = G7 for D♭7	**B𝄫7** = A7 for E♭7	**C♭7** for F7	**E𝄫7** = D7 for A♭7
A♭	**C♭7** for F7	**D♭7** for G7	**E𝄫7** = D7 for A♭7	**F♭7** for B♭7	**G♭7** for C7	**B𝄫7** = A7 for E♭7
E♭	**G♭7** for C7	**A♭7** for D7	**B𝄫7** = A7 for E♭7	**C♭7** for F7	**D♭7** for G7	**F♭7** for B♭7
B♭	**D♭7** for G7	**E♭7** for A7	**F♭7** for B♭7	**G♭7** for C7	**A♭7** for D7	**C♭7** for F7
F	**A♭7** for D7	**B♭7** for E7	**C♭7** for F7	**D♭7** for G7	**E♭7** for A7	**G♭7** for C7

C3 Modal interchange (borrowed chords)

C3 RULE Borrow from the PARALLEL modes (same tonic). Flat-side chords come from darker modes: ♭VI needs a ♭6, so Aeolian or darker.

Parallel modes share your tonic. Any chord from the degree triangle of a parallel mode can be borrowed. Darker modes give flat-side chords with a wistful, heavier colour; Lydian gives a brighter one. Using the brightness ladder (B1) you can find a borrowed chord's source: ♭VI needs a ♭6, so it comes from Aeolian or darker.

In a major key

Borrowed chord	Source (parallel)	Typical move
iv	Aeolian	IV–iv–I
♭VImaj7	Aeolian	♭VI–♭VII–I or ♭VI–I
♭VII7 (backdoor dominant)	Aeolian	Backdoor: iv7–♭VII7–Imaj7
♭IIImaj7	Aeolian or Dorian	I–♭III–IV
iiø7	Aeolian	iiø7–V7–Imaj7
v	Mixolydian	v–I7–IVmaj7
♭VIImaj7	Mixolydian or Dorian	I–♭VII–I
♭IImaj7	Phrygian	♭II–I
II7	Lydian	I–II7, not resolving to V

In a minor key

Borrowed chord	Source	Typical move
IV7	Dorian	i–IV7 vamp
V7	harmonic minor	V7–i
Imaj7	parallel major	Picardy ending
♭IImaj7	Phrygian	♭II–V–i or ♭II–i

Borrowed chords in all 12 major keys

Key	iv Aeolian	♭VImaj7 Aeolian	♭VII7 Aeolian	♭IIImaj7 Aeolian/Dorian	iiø7 Aeolian	v Mixolydian	♭VIImaj7 Mixolydian/Dorian	♭IImaj7 Phrygian	II7 Lydian
C	Fm7	A♭maj7	B♭7	E♭maj7	Dm7♭5	Gm7	B♭maj7	D♭maj7	D7
G	Cm7	E♭maj7	F7	B♭maj7	Am7♭5	Dm7	Fmaj7	A♭maj7	A7
D	Gm7	B♭maj7	C7	Fmaj7	Em7♭5	Am7	Cmaj7	E♭maj7	E7
A	Dm7	Fmaj7	G7	Cmaj7	Bm7♭5	Em7	Gmaj7	B♭maj7	B7
E	Am7	Cmaj7	D7	Gmaj7	F♯m7♭5	Bm7	Dmaj7	Fmaj7	F♯7
B	Em7	Gmaj7	A7	Dmaj7	C♯m7♭5	F♯m7	Amaj7	Cmaj7	C♯7
F♯	Bm7	Dmaj7	E7	Amaj7	G♯m7♭5	C♯m7	Emaj7	Gmaj7	G♯7
D♭	G♭m7	B𝄫maj7 = Amaj7	C♭7	F♭maj7	E♭m7♭5	A♭m7	C♭maj7	E𝄫maj7 = Dmaj7	E♭7
A♭	D♭m7	F♭maj7	G♭7	C♭maj7	B♭m7♭5	E♭m7	G♭maj7	B𝄫maj7 = Amaj7	B♭7
E♭	A♭m7	C♭maj7	D♭7	G♭maj7	Fm7♭5	B♭m7	D♭maj7	F♭maj7	F7
B♭	E♭m7	G♭maj7	A♭7	D♭maj7	Cm7♭5	Fm7	A♭maj7	C♭maj7	C7
F	B♭m7	D♭maj7	E♭7	A♭maj7	Gm7♭5	Cm7	E♭maj7	G♭maj7	G7

Borrowed chords in all 12 minor keys

Key	IV7 Dorian	V7 harmonic minor	Imaj7 parallel major	♭IImaj7 Phrygian
Am	D7	E7	Amaj7	B♭maj7
Em	A7	B7	Emaj7	Fmaj7
Bm	E7	F♯7	Bmaj7	Cmaj7
F♯m	B7	C♯7	F♯maj7	Gmaj7
C♯m	F♯7	G♯7	C♯maj7	Dmaj7
G♯m	C♯7	D♯7	G♯maj7	Amaj7
D♯m	G♯7	A♯7	D♯maj7	Emaj7
B♭m	E♭7	F7	B♭maj7	C♭maj7
Fm	B♭7	C7	Fmaj7	G♭maj7
Cm	F7	G7	Cmaj7	D♭maj7
Gm	C7	D7	Gmaj7	A♭maj7
Dm	G7	A7	Dmaj7	E♭maj7

D1 Guide tones of every 7th chord

D1 RULE The 3rd says major or minor; the 7th says major 7th, dominant or diminished. Only the dominant has a tritone between them.

The **3rd** says major or minor; the **7th** says major 7th, dominant or diminished. Together they identify every 7th-chord quality except one pair (m7 and m7♭5 share them; only the 5th tells them apart).

Among the four common 7th chords, **only the dominant has a tritone between its guide tones**. That tritone is the tension a V7 carries, and it's why the tritone sub works (C2).

Quality	3rd	7th	Between them	Formula
maj7 major 7	3	7	perfect 5th	1 3 5 7
7 dominant 7	3	♭7	tritone	1 3 5 ♭7
m7 minor 7	♭3	♭7	perfect 5th	1 ♭3 5 ♭7
m7♭5 half-diminished	♭3	♭7	perfect 5th	1 ♭3 ♭5 ♭7
mMaj7 minor-major 7	♭3	7	augmented 5th	1 ♭3 5 7
°7 diminished 7	♭3	𝄫7	tritone	1 ♭3 ♭5 𝄫7

Every 7th chord spelled on every root, guide tones in bold

Read across for one root, down for one quality. Both spellings are given for C♯/D♭ and F♯/G♭.

Root	maj7	7	m7	m7♭5	mMaj7	°7
C	C **E** G **B**	C **E** G **B♭**	C **E♭** G **B♭**	C **E♭** G♭ **B♭**	C **E♭** G **B**	C **E♭** G♭ **B𝄫**
C♯	C♯ **E♯** G♯ **B♯**	C♯ **E♯** G♯ **B**	C♯ **E** G♯ **B**	C♯ **E** G **B**	C♯ **E** G♯ **B♯**	C♯ **E** G **B♭**
D♭	D♭ **F** A♭ **C**	D♭ **F** A♭ **C♭**	D♭ **F♭** A♭ **C♭**	D♭ **F♭** A𝄫 **C♭**	D♭ **F♭** A♭ **C**	D♭ **F♭** A𝄫 **C𝄫**
D	D **F♯** A **C♯**	D **F♯** A **C**	D **F** A **C**	D **F** A♭ **C**	D **F** A **C♯**	D **F** A♭ **C♭**
E♭	E♭ **G** B♭ **D**	E♭ **G** B♭ **D♭**	E♭ **G♭** B♭ **D♭**	E♭ **G♭** B𝄫 **D♭**	E♭ **G♭** B♭ **D**	E♭ **G♭** B𝄫 **D𝄫**
E	E **G♯** B **D♯**	E **G♯** B **D**	E **G** B **D**	E **G** B♭ **D**	E **G** B **D♯**	E **G** B♭ **D♭**
F	F **A** C **E**	F **A** C **E♭**	F **A♭** C **E♭**	F **A♭** C♭ **E♭**	F **A♭** C **E**	F **A♭** C♭ **E𝄫**
F♯	F♯ **A♯** C♯ **E♯**	F♯ **A♯** C♯ **E**	F♯ **A** C♯ **E**	F♯ **A** C **E**	F♯ **A** C♯ **E♯**	F♯ **A** C **E♭**
G♭	G♭ **B♭** D♭ **F**	G♭ **B♭** D♭ **F♭**	G♭ **B𝄫** D♭ **F♭**	G♭ **B𝄫** D𝄫 **F♭**	G♭ **B𝄫** D♭ **F**	G♭ **B𝄫** D𝄫 **F𝄫**
G	G **B** D **F♯**	G **B** D **F**	G **B♭** D **F**	G **B♭** D♭ **F**	G **B♭** D **F♯**	G **B♭** D♭ **F♭**
A♭	A♭ **C** E♭ **G**	A♭ **C** E♭ **G♭**	A♭ **C♭** E♭ **G♭**	A♭ **C♭** E𝄫 **G♭**	A♭ **C♭** E♭ **G**	A♭ **C♭** E𝄫 **G𝄫**
A	A **C♯** E **G♯**	A **C♯** E **G**	A **C** E **G**	A **C** E♭ **G**	A **C** E **G♯**	A **C** E♭ **G♭**
B♭	B♭ **D** F **A**	B♭ **D** F **A♭**	B♭ **D♭** F **A♭**	B♭ **D♭** F♭ **A♭**	B♭ **D♭** F **A**	B♭ **D♭** F♭ **A𝄫**
B	B **D♯** F♯ **A♯**	B **D♯** F♯ **A**	B **D** F♯ **A**	B **D** F **A**	B **D** F♯ **A♯**	B **D** F **A♭**

Guide tones (3rd, 7th) of the seven diatonic chords in all 12 major keys

Key	Imaj7	ii7	iii7	IVmaj7	V7	vi7	viiø7
C	E B	F C	G D	A E	B F	C G	D A
G	B F♯	C G	D A	E B	F♯ C	G D	A E
D	F♯ C♯	G D	A E	B F♯	C♯ G	D A	E B
A	C♯ G♯	D A	E B	F♯ C♯	G♯ D	A E	B F♯
E	G♯ D♯	A E	B F♯	C♯ G♯	D♯ A	E B	F♯ C♯
B	D♯ A♯	E B	F♯ C♯	G♯ D♯	A♯ E	B F♯	C♯ G♯
F♯	A♯ E♯	B F♯	C♯ G♯	D♯ A♯	E♯ B	F♯ C♯	G♯ D♯
D♭	F C	G♭ D♭	A♭ E♭	B♭ F	C G♭	D♭ A♭	E♭ B♭
A♭	C G	D♭ A♭	E♭ B♭	F C	G D♭	A♭ E♭	B♭ F
E♭	G D	A♭ E♭	B♭ F	C G	D A♭	E♭ B♭	F C
B♭	D A	E♭ B♭	F C	G D	A E♭	B♭ F	C G
F	A E	B♭ F	C G	D A	E B♭	F C	G D

D2 The ii–V–I rule and the canonical lines in all 12 keys

D2 RULE Roots down a 5th: the 7th falls a step to the next 3rd; the 3rd holds to become the next 7th.

When roots move down a 5th, the guide tones barely move. **The 7th of the first chord falls a step to become the 3rd of the next. The 3rd of the first chord holds to become the 7th of the next.** So ii–V–I is two lines: at each change one note holds and the other steps down.

In C: Dm7 (F, C) → G7 (F, B) → Cmaj7 (E, B).

Derive any key this way: name the guide tones of the ii, then apply "7th falls, 3rd holds" twice.

D3 RULE 7th falls, 3rd holds (roots down a 5th). Tritone sub: same lines, only the root moves. Cycle of dominants: two chromatic lines falling. Blues I7–IV7: both fall a semitone.

1. The tritone sub leaves both lines exactly as they were. Only the root moves.
2. A cycle of dominants is two chromatic lines falling side by side, alternating 3rd and 7th.
3. In a blues, both guide tones fall a semitone from I7 to IV7 and rise back.

Line A starts on the 7th of the first chord (on the 3rd for the cycle and the blues); at every change each line moves to the nearest guide tone of the next chord. Minor ii–V–i is shown in the 12 minor keys.

Major ii–V–I

Key	Chords	Line A	Line B
C	Dm7 – G7 – Cmaj7	C → B → B	F → F → E
G	Am7 – D7 – Gmaj7	G → F♯ → F♯	C → C → B
D	Em7 – A7 – Dmaj7	D → C♯ → C♯	G → G → F♯
A	Bm7 – E7 – Amaj7	A → G♯ → G♯	D → D → C♯
E	F♯m7 – B7 – Emaj7	E → D♯ → D♯	A → A → G♯
B	C♯m7 – F♯7 – Bmaj7	B → A♯ → A♯	E → E → D♯
F♯	G♯m7 – C♯7 – F♯maj7	F♯ → E♯ → E♯	B → B → A♯
D♭	E♭m7 – A♭7 – D♭maj7	D♭ → C → C	G♭ → G♭ → F
A♭	B♭m7 – E♭7 – A♭maj7	A♭ → G → G	D♭ → D♭ → C
E♭	Fm7 – B♭7 – E♭maj7	E♭ → D → D	A♭ → A♭ → G
B♭	Cm7 – F7 – B♭maj7	B♭ → A → A	E♭ → E♭ → D
F	Gm7 – C7 – Fmaj7	F → E → E	B♭ → B♭ → A

Minor ii–V–i

Key	Chords	Line A	Line B
Am	Bm7♭5 – E7 – Am7	A → G♯ → G	D → D → C
Em	F♯m7♭5 – B7 – Em7	E → D♯ → D	A → A → G
Bm	C♯m7♭5 – F♯7 – Bm7	B → A♯ → A	E → E → D
F♯m	G♯m7♭5 – C♯7 – F♯m7	F♯ → E♯ → E	B → B → A
C♯m	D♯m7♭5 – G♯7 – C♯m7	C♯ → B♯ → B	F♯ → F♯ → E
G♯m	A♯m7♭5 – D♯7 – G♯m7	G♯ → F𝄪 → F♯	C♯ → C♯ → B
D♯m	E♯m7♭5 – A♯7 – D♯m7	D♯ → C𝄪 → C♯	G♯ → G♯ → F♯
B♭m	Cm7♭5 – F7 – B♭m7	B♭ → A → A♭	E♭ → E♭ → D♭
Fm	Gm7♭5 – C7 – Fm7	F → E → E♭	B♭ → B♭ → A♭
Cm	Dm7♭5 – G7 – Cm7	C → B → B♭	F → F → E♭
Gm	Am7♭5 – D7 – Gm7	G → F♯ → F	C → C → B♭
Dm	Em7♭5 – A7 – Dm7	D → C♯ → C	G → G → F

ii–subV–I

Key	Chords	Line A	Line B
C	Dm7 – D♭7 – Cmaj7	C → C♭ → B	F → F → E
G	Am7 – A♭7 – Gmaj7	G → G♭ → F♯	C → C → B
D	Em7 – E♭7 – Dmaj7	D → D♭ → C♯	G → G → F♯
A	Bm7 – B♭7 – Amaj7	A → A♭ → G♯	D → D → C♯
E	F♯m7 – F7 – Emaj7	E → E♭ → D♯	A → A → G♯
B	C♯m7 – C7 – Bmaj7	B → B♭ → A♯	E → E → D♯
F♯	G♯m7 – G7 – F♯maj7	F♯ → F → E♯	B → B → A♯
D♭	E♭m7 – E𝄫7 = D7 – D♭maj7	D♭ → D𝄫 → C	G♭ → G♭ → F
A♭	B♭m7 – B𝄫7 = A7 – A♭maj7	A♭ → A𝄫 → G	D♭ → D♭ → C
E♭	Fm7 – F♭7 – E♭maj7	E♭ → E𝄫 → D	A♭ → A♭ → G
B♭	Cm7 – C♭7 – B♭maj7	B♭ → B𝄫 → A	E♭ → E♭ → D
F	Gm7 – G♭7 – Fmaj7	F → F♭ → E	B♭ → B♭ → A

Blues I7–IV7

Key	Chords	Line A	Line B
C	C7 – F7	E → E♭	B♭ → A
G	G7 – C7	B → B♭	F → E
D	D7 – G7	F♯ → F	C → B
A	A7 – D7	C♯ → C	G → F♯
E	E7 – A7	G♯ → G	D → C♯
B	B7 – E7	D♯ → D	A → G♯
F♯	F♯7 – B7	A♯ → A	E → D♯
D♭	D♭7 – G♭7	F → F♭	C♭ → B♭
A♭	A♭7 – D♭7	C → C♭	G♭ → F
E♭	E♭7 – A♭7	G → G♭	D♭ → C
B♭	B♭7 – E♭7	D → D♭	A♭ → G
F	F7 – B♭7	A → A♭	E♭ → D

Cycle of dominants

Key	Chords	Line A	Line B
C	E7 – A7 – D7 – G7 – Cmaj7	G♯ → G → F♯ → F → E	D → C♯ → C → B → B
G	B7 – E7 – A7 – D7 – Gmaj7	D♯ → D → C♯ → C → B	A → G♯ → G → F♯ → F♯
D	F♯7 – B7 – E7 – A7 – Dmaj7	A♯ → A → G♯ → G → F♯	E → D♯ → D → C♯ → C♯
A	C♯7 – F♯7 – B7 – E7 – Amaj7	E♯ → E → D♯ → D → C♯	B → A♯ → A → G♯ → G♯
E	G♯7 – C♯7 – F♯7 – B7 – Emaj7	B♯ → B → A♯ → A → G♯	F♯ → E♯ → E → D♯ → D♯
B	D♯7 – G♯7 – C♯7 – F♯7 – Bmaj7	F𝄪 → F♯ → E♯ → E → D♯	C♯ → B♯ → B → A♯ → A♯
F♯	A♯7 – D♯7 – G♯7 – C♯7 – F♯maj7	C𝄪 → C♯ → B♯ → B → A♯	G♯ → F𝄪 → F♯ → E♯ → E♯
D♭	F7 – B♭7 – E♭7 – A♭7 – D♭maj7	A → A♭ → G → G♭ → F	E♭ → D → D♭ → C → C
A♭	C7 – F7 – B♭7 – E♭7 – A♭maj7	E → E♭ → D → D♭ → C	B♭ → A → A♭ → G → G
E♭	G7 – C7 – F7 – B♭7 – E♭maj7	B → B♭ → A → A♭ → G	F → E → E♭ → D → D
B♭	D7 – G7 – C7 – F7 – B♭maj7	F♯ → F → E → E♭ → D	C → B → B♭ → A → A
F	A7 – D7 – G7 – C7 – Fmaj7	C♯ → C → B → B♭ → A	G → F♯ → F → E → E

Turnaround from VI7

Key	Chords	Line A	Line B
C	A7 – Dm7 – G7 – Cmaj7	G → F → F → E	C♯ → C → B → B
G	E7 – Am7 – D7 – Gmaj7	D → C → C → B	G♯ → G → F♯ → F♯
D	B7 – Em7 – A7 – Dmaj7	A → G → G → F♯	D♯ → D → C♯ → C♯
A	F♯7 – Bm7 – E7 – Amaj7	E → D → D → C♯	A♯ → A → G♯ → G♯
E	C♯7 – F♯m7 – B7 – Emaj7	B → A → A → G♯	E♯ → E → D♯ → D♯
B	G♯7 – C♯m7 – F♯7 – Bmaj7	F♯ → E → E → D♯	B♯ → B → A♯ → A♯
F♯	D♯7 – G♯m7 – C♯7 – F♯maj7	C♯ → B → B → A♯	F𝄪 → F♯ → E♯ → E♯
D♭	B♭7 – E♭m7 – A♭7 – D♭maj7	A♭ → G♭ → G♭ → F	D → D♭ → C → C
A♭	F7 – B♭m7 – E♭7 – A♭maj7	E♭ → D♭ → D♭ → C	A → A♭ → G → G
E♭	C7 – Fm7 – B♭7 – E♭maj7	B♭ → A♭ → A♭ → G	E → E♭ → D → D
B♭	G7 – Cm7 – F7 – B♭maj7	F → E♭ → E♭ → D	B → B♭ → A → A
F	D7 – Gm7 – C7 – Fmaj7	C → B♭ → B♭ → A	F♯ → F → E → E

E1 The superimposition menus

E1 RULE Name the triad by the chord degree it sits on, then read each note as a tension over the chord root. Over a dominant: II = 9 #11 13, VI = 13 ♭9 3, ♭VI = ♭13 1 #9, ♭V = #11 ♭7 ♭9.

E2 RULE Pentatonics a step or a 5th up stay inside (Dorian, Lydian, Mixolydian colours); the minor pentatonic a minor 3rd above a dominant is the altered sound.

A triad played over a chord is a **superimposition**: three notes you already know as a shape, heard as **tensions** of the chord underneath. Name it by the chord degree it is built on, not by its letter, and it transposes to every key at once: over any dominant, the **II** major triad gives **9, #11, 13**.

To work one out: count up from the chord root to the triad root, then read each triad note as an interval from the chord root. D over C7: D is the 9, F# the #11, A the 13.

Why superimpositions work

1. **They are the chord's upper floors.** Stack any chord in thirds past the 7th: 1 3 5 7 9 11 13. A triad built on the 5th, 7th or 9th of the chord is three neighbouring floors of that stack, so it isn't a foreign chord at all: it is the top of a bigger version of the same chord (D over C7 = the top of C13#11).
2. **They come from the chord's own scale.** Each option is a subset of the scale that fits the chord (Lydian dominant, altered, Dorian…). Pick the structure and you have picked the colour.
3. **They skip the avoid notes.** The note a half step above a chord tone (the 11 over a major or dominant chord) rubs on a long note. Good superimpositions leave it out, or replace it with #11.
4. **The ear trusts strong shapes.** A triad or a pentatonic is so familiar that even far-out notes sound deliberate: you are playing a clear idea, not wrong notes.
5. **Tension has somewhere to go.** Over a dominant, altered notes sit a half step from notes of the I chord, so they pull into it when the chord resolves.

Worked example, G♭ B♭ D♭ over C7 = ♯11, ♭7, ♭9.

- **Upper floors.** Stack C Altered in thirds: C E♭ G♭ B♭ D♭ F♭ A♭ (1 ♯9 ♯11 ♭7 ♭9 3 ♭13). The G♭ triad is three neighbouring floors of that stack: the upper part of the full chord, with the bass and comping supplying the lower notes.
- **From the chord's own scale.** Every note is in C Altered (C D♭ E♭ F♭ G♭ A♭ B♭), the scale that fits C7, so it adds colour without fighting the chord.
- **Guide tones.** It contains the 7th, so it still names the chord's quality; the other notes are colour.
- **Resolving to F.** G♭ rises to G (9); B♭ falls to A (3); D♭ falls to C (5). Half-step pulls like these are what make the tension sound intentional: it has somewhere to go.
- **Sound.** G♭ over C7: ♯11, ♭7, ♭9 — the triad of the tritone sub (G♭7), so it bridges C7 and its substitute.

A pentatonic has no half steps, so it never clashes hard. Played from other roots it becomes a five-note colour over the chord: a step above a minor 7 gives Dorian (9 11 5 13 1); a minor 3rd above a dominant gives five altered notes (♯9 ♯11 ♭13 ♭7 ♭9).

Worked example, E♭ G♭ A♭ B♭ D♭ over C7 = ♯9, ♯11, ♭13, ♭7, ♭9.

- **From the chord's own scale.** Every note is in C Altered (C D♭ E♭ F♭ G♭ A♭ B♭), the scale that fits C7, so it adds colour without fighting the chord.
- **Guide tones.** It contains the 7th, so it still names the chord's quality; the other notes are colour.
- **Resolving to F.** E♭ rises to E (7); G♭ rises to G (9); A♭ rises to A (3); B♭ falls to A (3); D♭ falls to C (5). Half-step pulls like these are what make the tension sound intentional: it has somewhere to go.
- **Sound.** Minor pentatonic a minor 3rd up: ♯9, ♯11, ♭13, ♭7, ♭9 — five altered notes, no root, no 3rd.

Over a major 7 (Xmaj7)

Play	Adds	Sounds like	Scale	Why
V major triad	5 7 9	maj9	Ionian	The triad a 5th up adds the 7 and 9 on top of the chord: the plainest, prettiest upper structure.
II major triad	9 ♯11 13	maj13♯11	Lydian	The major triad a step up gives 9, ♯11 and 13: the whole Lydian colour in three notes.
iii minor triad	3 5 7	maj7 (no root)	Ionian	The minor triad on the 3rd is the chord without its root: 3, 5, 7. Safe, inside, and a smooth way to target the 3rd.
vii minor triad	7 9 ♯11	maj9♯11	Lydian	B minor over C: 7, 9, ♯11. Bright and floating.
II major pentatonic	9 3 ♯11 13 7	maj13♯11	Lydian	Major pentatonic a step up: 9, 3, ♯11, 13, 7 — no root and no 4th, so nothing clashes.
V major pentatonic	5 13 7 9 3	maj13	Ionian	Major pentatonic a 5th up: 5, 13, 7, 9, 3 — Ionian without the avoid note (the 4th).

Over a dominant 7 (X7)

Play	Adds	Sounds like	Scale	Why
v minor triad	5 ♭7 9	9	Mixolydian	The minor triad a 5th up: 5, ♭7, 9. Plain, inside dominant sound.
II major triad	9 ♯11 13	13♯11	Lydian dominant	The major triad a step up: 9, ♯11, 13. The Lydian dominant sound used on IV7, ♭VII7 and tritone subs.
VI major triad	13 ♭9 3	13♭9	half-whole diminished	The major triad a major 6th up: 13, ♭9, 3. Bright and tense at once; resolves beautifully.
♭III major triad	♯9 5 ♭7	7♯9	half-whole diminished	E♭ over C7: ♯9, 5, ♭7. The bluesy ♯9 sound.
♭V major triad	♯11 ♭7 ♭9	7♭9♯11	altered / half-whole	G♭ over C7: ♯11, ♭7, ♭9 — the triad of the tritone sub (G♭7), so it bridges C7 and its substitute.
♭VI major triad	♭13 1 ♯9	7♯9♭13	altered	A♭ over C7: ♭13, 1, ♯9. Dark and altered, resolves down to the I.
♭iii minor pentatonic	♯9 ♯11 ♭13 ♭7 ♭9	7alt	altered	Minor pentatonic a minor 3rd up: ♯9, ♯11, ♭13, ♭7, ♭9 — five altered notes, no root, no 3rd.
v minor pentatonic	5 ♭7 1 9 11	9sus4	Mixolydian	Minor pentatonic a 5th up: 5, ♭7, 1, 9, 11. Suspended, open, great on 7sus4.

Over a minor 7 (Xm7)

Play	Adds	Sounds like	Scale	Why
♭III major triad	♭3 5 ♭7	m7 (no root)	Dorian / Aeolian	E♭ over Cm7: ♭3, 5, ♭7. The chord without its root.
v minor triad	5 ♭7 9	m9	Dorian / Aeolian	The minor triad a 5th up: 5, ♭7, 9.
♭VII major triad	♭7 9 11	m11	Dorian / Aeolian	B♭ over Cm7: ♭7, 9, 11. Open, modern minor.
IV major triad	11 13 1	m13	Dorian	F over Cm7: 11, 13, 1. The natural 13 is the Dorian note.
i minor pentatonic	1 ♭3 11 5 ♭7	m11	Dorian / Aeolian	The home minor pentatonic: 1, ♭3, 11, 5, ♭7.
ii minor pentatonic	9 11 5 13 1	m13	Dorian	Minor pentatonic a step up: 9, 11, 5, 13, 1 — the Dorian pentatonic.
v minor pentatonic	5 ♭7 1 9 11	open m11 (no 3rd)	Dorian / Aeolian	Minor pentatonic a 5th up: 5, ♭7, 1, 9, 11 — no 3rd, so it floats.

Over a half-diminished (Xm7♭5)

Play	Adds	Sounds like	Scale	Why
♭iii minor triad	♭3 ♭5 ♭7	m7♭5 (no root)	Locrian	E♭m over Cø7: ♭3, ♭5, ♭7. The chord without its root.
♭VI major triad	♭13 1 ♭3	m7♭5♭13	Locrian	A♭ over Cø7: ♭13, 1, ♭3.
♭VII major triad	♭7 9 11	m9♭5	Locrian ♮2	B♭ over Cø7: ♭7, 9, 11. The natural 9 is the melodic-minor (Locrian ♮2) colour.
iv minor pentatonic	11 ♭13 ♭7 1 ♭3	m11♭5	Locrian	Minor pentatonic a 4th up: 11, ♭13, ♭7, 1, ♭3.
♭iii minor pentatonic	♭3 ♭5 ♭13 ♭7 ♭9	m7♭5♭9	Locrian	Minor pentatonic a minor 3rd up: ♭3, ♭5, ♭13, ♭7, ♭9.

Over a minor-major 7 / minor 6 (Xm(maj7))

Play	Adds	Sounds like	Scale	Why
V major triad	5 7 9	m(maj9)	melodic minor	G over Cm: 5, 7, 9. The major triad a 5th up is the sound of melodic minor.
IV major triad	11 13 1	m6/9	melodic minor / Dorian	F over Cm: 11, 13, 1.
ii minor pentatonic	9 11 5 13 1	m6/9	melodic minor	Minor pentatonic a step up: 9, 11, 5, 13, 1.

Over a diminished 7 (X°7)

Play	Adds	Sounds like	Scale	Why
II major triad	9 ♭5 °7	°7 with 9, 11	whole-half diminished	D over C°7: 9, ♭5, 13(=°7). Any major triad a step above a dim7 note fits.
VII major triad	7 ♭3 ♭5	°(maj7)	whole-half diminished	B over C°7: 7, ♭3, ♭5. The leading-tone triad hidden in the scale.

E2 Superimposition in all 12 keys

The structure to play on every root, for the menu items worth knowing instantly. Header: the degree, the tensions it adds and the resulting chord. Structures are spelled the simplest way (they are played as shapes), so the enharmonic of the chord root is used where it avoids double accidentals.

Over a dominant 7th

Chord	II major triad 9 ♯11 13 → 13♯11	VI major triad 13 ♭9 3 → 13♭9	♭VI major triad ♭13 1 ♯9 → 7♯9♭13	♭V major triad ♯11 ♭7 ♭9 → 7♭9♯11	♭iii minor pent. ♯9 ♯11 ♭13 ♭7 ♭9 → 7alt
C7	**D** D F♯ A	**A** A C♯ E	**A♭** A♭ C E♭	**G♭** G♭ B♭ D♭	**E♭m pent.** E♭ G♭ A♭ B♭ D♭
G7	**A** A C♯ E	**E** E G♯ B	**E♭** E♭ G B♭	**D♭** D♭ F A♭	**B♭m pent.** B♭ D♭ E♭ F A♭
D7	**E** E G♯ B	**B** B D♯ F♯	**B♭** B♭ D F	**A♭** A♭ C E♭	**Fm pent.** F A♭ B♭ C E♭
A7	**B** B D♯ F♯	**F♯** F♯ A♯ C♯	**F** F A C	**E♭** E♭ G B♭	**Cm pent.** C E♭ F G B♭
E7	**F♯** F♯ A♯ C♯	**D♭** D♭ F A♭	**C** C E G	**B♭** B♭ D F	**Gm pent.** G B♭ C D F
B7	**D♭** D♭ F A♭	**A♭** A♭ C E♭	**G** G B D	**F** F A C	**Dm pent.** D F G A C
F♯7 / G♭7	**A♭** A♭ C E♭	**E♭** E♭ G B♭	**D** D F♯ A	**C** C E G	**Am pent.** A C D E G
D♭7 / C♯7	**E♭** E♭ G B♭	**B♭** B♭ D F	**A** A C♯ E	**G** G B D	**Em pent.** E G A B D
A♭7 / G♯7	**B♭** B♭ D F	**F** F A C	**E** E G♯ B	**D** D F♯ A	**Bm pent.** B D E F♯ A
E♭7 / D♯7	**F** F A C	**C** C E G	**B** B D♯ F♯	**A** A C♯ E	**F♯m pent.** F♯ A B C♯ E
B♭7 / A♯7	**C** C E G	**G** G B D	**G♭** G♭ B♭ D♭	**E** E G♯ B	**C♯m pent.** C♯ E F♯ G♯ B
F7	**G** G B D	**D** D F♯ A	**D♭** D♭ F A♭	**B** B D♯ F♯	**G♯m pent.** G♯ B C♯ D♯ F♯

Over a major 7th

Chord	II major triad 9 ♯11 13 → maj13♯11	V major triad 5 7 9 → maj9	II major pent. 9 3 ♯11 13 7 → maj13♯11	V major pent. 5 13 7 9 3 → maj13
Cmaj7	**D** D F♯ A	**G** G B D	**D pent.** D E F♯ A B	**G pent.** G A B D E
Gmaj7	**A** A C♯ E	**D** D F♯ A	**A pent.** A B C♯ E F♯	**D pent.** D E F♯ A B
Dmaj7	**E** E G♯ B	**A** A C♯ E	**E pent.** E F♯ G♯ B C♯	**A pent.** A B C♯ E F♯
Amaj7	**B** B D♯ F♯	**E** E G♯ B	**B pent.** B C♯ D♯ F♯ G♯	**E pent.** E F♯ G♯ B C♯
Emaj7	**F♯** F♯ A♯ C♯	**B** B D♯ F♯	**F♯ pent.** F♯ G♯ A♯ C♯ D♯	**B pent.** B C♯ D♯ F♯ G♯
Bmaj7	**D♭** D♭ F A♭	**F♯** F♯ A♯ C♯	**D♭ pent.** D♭ E♭ F A♭ B♭	**F♯ pent.** F♯ G♯ A♯ C♯ D♯
F♯maj7 / G♭maj7	**A♭** A♭ C E♭	**D♭** D♭ F A♭	**A♭ pent.** A♭ B♭ C E♭ F	**D♭ pent.** D♭ E♭ F A♭ B♭
D♭maj7 / C♯maj7	**E♭** E♭ G B♭	**A♭** A♭ C E♭	**E♭ pent.** E♭ F G B♭ C	**A♭ pent.** A♭ B♭ C E♭ F
A♭maj7 / G♯maj7	**B♭** B♭ D F	**E♭** E♭ G B♭	**B♭ pent.** B♭ C D F G	**E♭ pent.** E♭ F G B♭ C
E♭maj7 / D♯maj7	**F** F A C	**B♭** B♭ D F	**F pent.** F G A C D	**B♭ pent.** B♭ C D F G
B♭maj7 / A♯maj7	**C** C E G	**F** F A C	**C pent.** C D E G A	**F pent.** F G A C D
Fmaj7	**G** G B D	**C** C E G	**G pent.** G A B D E	**C pent.** C D E G A

Over a minor 7th

Chord	♭VII major triad ♭7 9 11 → m11	IV major triad 11 13 1 → m13	ii minor pent. 9 11 5 13 1 → m13	v minor triad 5 ♭7 9 → m9
Cm7	**B♭** B♭ D F	**F** F A C	**Dm pent.** D F G A C	**Gm** G B♭ D
Gm7	**F** F A C	**C** C E G	**Am pent.** A C D E G	**Dm** D F A
Dm7	**C** C E G	**G** G B D	**Em pent.** E G A B D	**Am** A C E
Am7	**G** G B D	**D** D F♯ A	**Bm pent.** B D E F♯ A	**Em** E G B
Em7	**D** D F♯ A	**A** A C♯ E	**F♯m pent.** F♯ A B C♯ E	**Bm** B D F♯
Bm7	**A** A C♯ E	**E** E G♯ B	**C♯m pent.** C♯ E F♯ G♯ B	**F♯m** F♯ A C♯
F♯m7 / G♭m7	**E** E G♯ B	**B** B D♯ F♯	**G♯m pent.** G♯ B C♯ D♯ F♯	**C♯m** C♯ E G♯
D♭m7 / C♯m7	**B** B D♯ F♯	**F♯** F♯ A♯ C♯	**D♯m pent.** D♯ F♯ G♯ A♯ C♯	**G♯m** G♯ B D♯
A♭m7 / G♯m7	**G♭** G♭ B♭ D♭	**D♭** D♭ F A♭	**B♭m pent.** B♭ D♭ E♭ F A♭	**E♭m** E♭ G♭ B♭
E♭m7 / D♯m7	**D♭** D♭ F A♭	**A♭** A♭ C E♭	**Fm pent.** F A♭ B♭ C E♭	**B♭m** B♭ D♭ F
B♭m7 / A♯m7	**A♭** A♭ C E♭	**E♭** E♭ G B♭	**Cm pent.** C E♭ F G B♭	**Fm** F A♭ C
Fm7	**E♭** E♭ G B♭	**B♭** B♭ D F	**Gm pent.** G B♭ C D F	**Cm** C E♭ G

Say it as a sentence: “over G7, A is 13♯11, E is 13♭9, E♭ is 7♯9♭13, D♭ is 7♭9♯11, B♭ minor pentatonic is 7alt.” Then move the chord round the circle and say it again.

Daily recall checklist

Be able to reel these off with no instrument and no paper. Tick a line when it is instant in all 12 keys.

A · Keys

- ☐ Semitones of any interval, its inversion, and 9/11/13 as 2/4/6 — instantly.
- ☐ The circle both ways, the sharps and flats of any key in order, and the key from any signature (both shortcuts).
- ☐ Relative and parallel minor of every major key; relative major of every minor key.
- ☐ Major, natural, harmonic and melodic minor of today's key, aloud, in under 5 seconds each.
- ☐ The seven 7th chords of any major and any minor key, with the mode on each degree.
- ☐ Where a chord lives: every major key a m7, maj7, 7 or m7♭5 belongs to.

B · Scales and modes

- ☐ The brightness ladder with the changed note on each rung, and the parent of each major mode.
- ☐ Any major mode on any root, by the ladder and then checked by the parent.
- ☐ Harmonic- and melodic-minor modes: name, twin + change, 7th chord and use, in order.
- ☐ The parent shortcuts in both directions (X7alt → melodic minor a semitone up, and back).
- ☐ Which four dominants share each half-whole scale; the two whole-tone sets.
- ☐ First-choice scale for every chord quality and every dominant job.

C · Functional harmony

- ☐ The five secondary dominants of any major key with their out-of-key notes; the minor-key set.
- ☐ The six tritone pairs and the guide tones they share; subV7 of any target.
- ☐ The borrowed chords of any major key with their source modes; the backdoor ♭VII7.

D · Guide tones

- ☐ 3rd and 7th of maj7, 7, m7, m7♭5, mMaj7 and °7 on any root.
- ☐ "7th falls, 3rd holds": both lines of ii–V–I in all 12 keys, then minor ii–V–i, ii–subV–I.
- ☐ The cycle of dominants, the turnaround from VI7 and the blues I7–IV7 lines in today's key.

E · Superimposition

- ☐ Over any dominant: II, VI, ♭VI, ♭V triads and the ♭iii minor pentatonic — name each one and the sound.
- ☐ Over any maj7: II and V triads; over any m7: ♭VII and IV triads and the ii minor pentatonic.
- ☐ Read any triad over any root as tensions (D over C7 = 9 ♯11 13).

The daily routine (10 minutes): pick today's key by walking the circle (C, G, D, A, E, B, F♯, D♭, A♭, E♭, B♭, F). Say its signature and scales, its seven chords, its secondary dominants and borrowed chords, and the ii–V–I guide-tone lines. Then name the superimpositions over its V7. Anything slower than a second goes on tomorrow's list.